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MASTER'S THESIS

Design of Switch-Mode Converter for Battery Emulator Power Supply

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ABSTRACT

IPPSuM, a battery emulator power supply, has been designed for testing battery-powered devices. The power transfer of the compact power supply is achieved with a DC-DC switch-mode converter, followed by a linear regulator. The problem with this particular power supply is its relatively low output voltage and power, which limits the number of devices that can be tested with it. Therefore, in this thesis, the output voltage and power ranges of the DC-DC switch-mode converter were increased. This was the first necessary step to upgrade the entire power supply in the future.

The starting point of the design was to achieve all the other requirements of the old power supply. The design of the DC-DC converter started with the selection of topology, in which case the active clamp forward converter was chosen as the suitable topology. After the topology selection, the components were sized according to the constraints of the specification. Finally, this was followed by the implementation of the simulation model and typical simulations for a DC-DC converter with the Tina-Ti software.

The output voltage range of the old DC-DC converter was measured to be 5.18 V – 11.45 V, and the goal was to increase this range to 5.5 V – 21.5 V. The simulation results showed that the output voltage range was increased relatively successfully, being 5.75 V – 21.58 V after the upgrade. The designed DC-DC converter also performed well in terms of other requirements. The minimum input voltage simulation was the only exception as the output voltage failed to reach the required level. To solve this, the turns ratio of the DC-DC converter’s power transformer would have to be increased and the relevant components resized. In addition to the good simulation results, the feedback circuit was significantly simplified. This lowers the cost of the converter while allowing more circuit board space.

As stated earlier, the simulations covered only the most typical DC-DC converter simulations. More versatile and thorough simulations would be needed in order to ensure the performance of the DC-DC converter in all situations.

Key words: DC-DC, active clamp forward

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TIIVISTELMÄ

IPPSuM, akkuemulaattoriteholähde on suunniteltu akkukäyttöisten laitteiden testaamiseen. Pienikokoisen teholähteen tehonsiirto on toteutettu DC-DC hakkurimuuntimella, jota seuraa lineaariregulaattori. Teholähteen ongelma on sen suhteellisen pieni lähtöjännite ja -teho, mikä rajoittaa sillä testattavien laitteiden määrää. Tämän vuoksi, tässä diplomityössä suurennettiin hakkurimuuntimen lähtöjännite- ja -tehoaluetta. Tämä oli ensimmäinen vaadittava askel, jotta koko teholähde voidaan päivittää tulevaisuudessa.

Suunnittelun lähtökohta oli saavuttaa kaikki muut vanhan teholähteen vaatimusmäärittelyt. Hakkurimuuntimen suunnittelu alkoi topologian valinnalla, jolloin sopivaksi muuntimeksi valikoitui active clamp forward -muunnin. Tämän jälkeen komponentit voitiin mitoittaa vaatimusmäärittelyn reunaehtojen mukaisesti. Tätä seurasi lopuksi simulointimallin toteuttaminen ja hakkurimuuntimelle tyypilliset simuloinnit Tina-Ti -ohjelmistolla.

Vanhan hakkurimuuntimen lähtöjännitealue mitattiin olevan 5.18 V – 11.45 V ja tavoite oli nostaa tämä alue 5.5 V – 21.5 V alueelle. Simulointitulokset näyttivät, että lähtöjännitealue saatiin suurennettua suhteellisen onnistuneesti sen ollessa päivityksen jälkeen 5.75 V – 21.58 V. Suunniteltu hakkurimuunnin suoriutui myös muista vaatimusmäärittelyistä hyvin, lukuunottamatta pienimmällä vaaditulla tulojännitteellä tehtyä simulointia, missä lähtöjännite ei saavuttanut riittävää tasoa. Tämän ratkaisemiseksi, hakkurimuuntimen tehomuuntaja täytyisi vaihtaa kierrossuhteeltaan isompaan ja mitoittaa siihen liittyvät komponentit uudestaan. Hyvien simulointituloksien lisäksi takaisinkytkentäpiiriä saatiin yksinkertaistettua huomattavasti. Tämän ansiosta saadaan säästettyä sekä kustannuksissa, että piirilevytilassa.

Kuten aiemmin todettiin, simuloinnit kattoivat vain tyypillisimmät hakkurimuuntimen simuloinnit. Monipuolisemmat ja perusteellisemmat simuloinnit tarvittaisiin, jotta hakkurimuuntimen käyttäytyminen voitaisiin varmistaa kaikissa tilanteissa.

Avainsanat: DC-DC, active clamp forward

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FOREWORD

This master's thesis was carried out in GPV Oulu, and the purpose was to increase the output voltage and power ranges of the DC-DC converter of the IPPSuM battery emulator power supply.

First, I want to thank my former manager, Risto Mäkelä, who hired me as a summer trainee and gave me an opportunity to start this thesis work. I would also like to express my gratitude to Jari Korhonen, who was also my former manager/director and allowed me to finish this thesis despite the challenging and uncertain circumstances. I am very grateful to my technical advisor, D.Sc. Ari Kilpelä, for his excellent guidance and close support during this thesis.

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Teemu Saukkosaari

LIST OF ABBREVIATIONS AND SYMBOLS

AC	Alternating Current
BCM	Boundary or Borderline Conduction Mode
CCM	Continuous Conduction Mode
CRM	Critical Conduction Mode
CTR	Current Transfer Ratio
DC	Direct Current
DCM	Discontinuous Conduction Mode
EMI	Electromagnetic Interference
ESR	Equivalent Series Resistance
LAN	Local Area Network
LED	Light Emitting Diode
MCU	Microcontroller Unit
MOSFET	Metal-Oxide-Semiconductor Field-Effect Transistor
PCB	Printed Circuit Board
PWM	Pulse-Width Modulation
RMS	Root Mean Square
SPICE	Simulation Program with Integrated Circuit Emphasis
USB	Universal Serial Bus
ZSF	Zero-Scale Factor
ZVS	Zero Voltage Switching

A	*Ampere*
A_{AVG}	*Average Current in Amperes*
A_{PP}	*Peak-to-Peak Current in Amperes*
A_{RMS}	*Root Mean Square Current in Amperes*
C	*Capacitance*
C_{CLAMP}	*Capacitance of The Clamp Capacitor*
C_F	*Capacitance of The C_F Capacitor*
C_O	*Output Capacitance*
$CP1$	*Capacitance of The CP1 Capacitor*
$CZ2$	*Capacitance of The CZ2 Capacitor*
$CZ3$	*Capacitance of The CZ3 Capacitor*
D	*Duty Cycle*
D_{MAX}	*Maximum Duty Cycle*
D_{MIN}	*Minimum Duty Cycle*
dV_L/dt	*Voltage Equivalent Compensation Ramp*
fc	*Cross-Over Frequency*
F_{SW}	*Switching Frequency*
I_{IN}	*Input Current*
$I_{LO(PK)}$	*Secondary Side Peak Current*
$I_{LO(RMS)}$	*RMS Inductor Current*
I_{MAG}	*Magnetizing Current*
I_{OPTO}	*LED Current*
$I_{OPTO(MAX)}$	*Maximum LED Current*
I_{OUT}	*Output Current*
$I_{PRI(PK)}$	*Primary Side Peak Current*

I_{REF}	*Reference Current*
$I_{STEP(MAX)}$	*Maximum Value of Step Load*
$I_{STEP(MIN)}$	*Minimum Value of Step Load*
L	*Inductance*
L_{MAG}	*Magnetizing Inductance of The Transformer*
$L_{MAG(P)}$	*Magnetizing Inductance of The Transformer on The Primary Side*
$L_{MAG(S)}$	*Magnetizing Inductance of The Transformer on The Secondary Side*
L_O	*Output Inductance*
L_S	*Secondary Inductance of The Transformer*
$L_{M,crit}$	*Boundary Value of Magnetizing Inductance*
m	*Level of Desired Slope Compensation*
N_{CS}	*Current Sense Transformer Turns Ratio*
N_p	*Number of The Primary Turns*
N_s	*Number of The Secondary Turns*
P	*Power*
P_{IN}	*Input Power*
P_{OUT}	*Output Power*
R	*Resistance*
R_{CS}	*Resistance of The* R_{CS} *Resistor*
R_{DEL}	*Resistance of The* R_{DEL} *Resistor*
$R_{ESR(MAX)}$	*Maximum Equivalent Series Resistance*
R_F	*Resistance of The* R_F *Resistor*
R_L	*Load Resistance*
R_{OPTO}	*Resistance of The* R_{OPTO} *Resistor*
R_{SLOPE}	*Resistance of The* R_{SLOPE} *Resistor*
R_{TOFF}	*Resistance of The* R_{TOFF} *Resistor*
R_{TON}	*Resistance of The* R_{TON} *Resistor*
R_{VREF}	*Resistance of The* R_{VREF} *Resistor*
RZ2	*Resistance of The RZ2 Resistor*
RZ3	*Resistance of The RZ3 Resistor*
t_{DEL1}	*Delay Time (AUX to OUT)*
t_{DEL2}	*Delay Time (OUT to AUX)*
t_{OFF}	*Off-Time of The Switch*
t_{ON}	*On-Time of The Switch*
T_{sw}	*Switching Period*
V	*Volt*
V_{CL}	*Clamp Voltage*
$V_{CL(HS)}$	*High-Side Clamp Voltage*
$V_{CL(LS)}$	*Low-Side Clamp Voltage*
V_{CS}	*Current Sense Threshold Voltage*
$V_{CS(MIN)}$	*Minimum Control Signal Value in Volts*
VDC	*Volts Direct Current*
V_{DS}	*Drain-to-Source Voltage*
$V_{DS(Q1)}$	*Drain-to-Source Voltage of The Switch* Q_1
$V_{DS(QMAIN)}$	*Drain-to-Source Voltage of The Switch QMAIN*
V_F	*Forward Voltage*
$V_{FB(MIN)}$	*Minimum Feedback Voltage*
V_{IN}	*Input Voltage*

$V_{IN(MIN)}$	*Minimum Input Voltage*
V_{OPTO}	*Supply Voltage of The Optocoupler*
$V_{OS(MAX)}$	*Maximum Overshoot Voltage*
$V_{OS(MIN)}$	*Maximum Stabile Output Voltage*
V_{OUT}	*Output Voltage*
$V_{OUT(MAX)}$	*Maximum Output Voltage*
V_{RAMP}	*Amplitude of The Ramp Signal Voltage*
V_{REF}	*Reference Voltage*
V_{RESET}	*Primary Voltage During Transformer Reset*
ΔI_{LO}	*The Maximum Ripple Current of The Output Inductor*
$\Delta I_{LO(\%)}$	The *Maximum Allowed Ripple Current in Percents*
ΔV_{CL}	*Difference in The Clamp Voltages*
$\Delta V_{OUT(RIP)}$	*Output Ripple Voltage*

μ	Micro
η	Power Efficiency
Ω	Ohm

1 INTRODUCTION

Modern electrical and rechargeable devices often have charger circuits integrated right into them. When testing these devices, a battery emulator power supply is beneficial since it can act as a battery and provide the required power for the device. By changing and varying the input voltage of the device, its behavior can be monitored in different operating conditions. Another important feature of the battery emulator is its ability to sink current. When the emulator is able to receive current, the properties of the charger may be measured and tested. Examples of these properties are stable constant current, which the emulator can measure directly, and automatic turn-off, which can be tested by varying the emulator's voltage and current. Instead of using an actual battery, the battery emulator power supply saves time as the charging and discharging of the battery can be avoided. Therefore, research and development as well as production testing benefit from the battery emulators.

The IPPSuM 1.0 power supply unit is a product of GPV, which is a company that provides global electronic manufacturing services. The product was developed by PKC Electronics in 2012, before the company was sold to Enics in 2017. In 2022, Enics merged with GPV, taking ownership of the product. The product is designed for testing battery-powered devices, for example, mobile phones and smart watches, as well as their chargers. The overview of the IPPSuM 1.0 is shown in Figure 1. As can be seen, it requires direct current (DC) voltage from 11 V to 13.5 V as an input voltage from an external source and it has two identical isolated outputs, which are capable of providing 0 to 10 V and 0 to 5 A. However, the maximum output power is limited to 30 W. The advantages of the product are its compact size, accurate low-current measurement, and current sink capability up to 1 A.

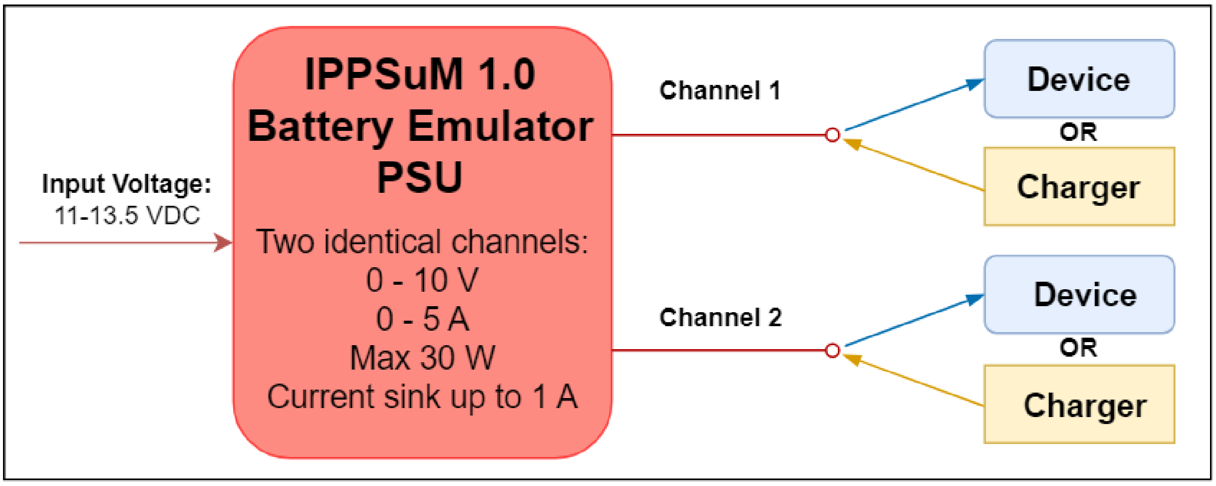


Figure 1. Overview of the IPPSuM 1.0 battery emulator power supply unit.

Due to the age of the product, it has a lot of end-of-life components, and it is missing modern industry features, such as a Universal Serial Bus (USB) Type-C connection, which is frequently used by a wide variety of modern devices. Furthermore, the range of devices that can be tested with the old IPPSuM 1.0 is limited by its relatively low output voltage and power. This range can be extended by increasing the maximum output voltage and power of the IPPSuM. This also justifies the implementation of the USB Type-C, which would further increase the number of possible devices to be tested.

The IPPSuM 1.0 utilizes an isolated switch-mode converter followed by a linear regulator to achieve its power transfer. In order to increase the overall output voltage and power ranges, upgrading the switch-mode stage should be the first step. Therefore, the goal of this thesis is to design and upgrade the old IPPSuM 1.0's isolated DC-DC switch-mode converter to operate with wider output voltage and power ranges. This design should also meet the existing requirements and specifications of the old converter. The new IPPSuM 2.0 would be able to be designed completely after the DC-DC converter is upgraded, increasing the number of possible devices that can be tested with it in the future. The design work includes component sizing, simulation model creation, and typical converter simulations of the final design, giving a good basis for the upgrade.

This thesis first introduces the basic theory of DC-DC switch-mode converters. The converter topologies that are found suitable for the new design are discussed more in detail. After the theory part, the IPPSuM and the new converter requirements are presented, followed by the topology selection and component sizing calculations. After the component sizing, the performance of the converter is simulated using Tina-Ti software. At the end, the simulation results are discussed and summarized.

2 DC-DC SWITCH-MODE POWER SUPPLY TOPOLOGIES

Most of the power in the grid today is transmitted as alternating current (AC) because it is easy to step up or down [1]. However, many devices and appliances cannot directly use the AC power delivered. The power must be converted to DC for various electronic appliances. In addition, the operating voltage of the appliances varies. These factors make energy conversion necessary in multiple ways: AC-DC conversion, DC-DC conversion, or DC-AC conversion.

A conversion is accomplished by using a converter, which takes an input voltage and outputs a different voltage. Various operating circumstances, such as increased load current, can affect the output voltage level of the converter. The converter requires a voltage regulator in order to produce a stable output voltage under all working conditions. However, the term converter is often used for a circuit that also regulates the output voltage. [2]

The two primary types of DC-DC converters and regulators are switch-mode and linear mode, each with their own benefits and disadvantages. Furthermore, the switching regulators can be divided into isolated and non-isolated subcategories. Voltage step-down and step-up converters buck and boost, as well as their combination buck-boost, are the most basic non-isolated converters. These converters serve as the basis for several isolated converter topologies. For example, a flyback is an isolated converter, which is basically a buck-boost converter using a transformer instead of an inductor. [2]

Galvanic isolation is frequently used in a wide range of applications, not only for safety reasons and to achieve various ground references but also to break ground loops that could generate unwanted voltages and currents. Galvanic isolation separates the input and output sides of a circuit in such a way that there is no direct conduction path for the current. A typical way to establish isolation in converters is to use transformers, which transfer the power from the primary side to the secondary side by using electromagnetic induction. Optocouplers are another method of providing isolation. In switching regulators, they are typically used in feedback loops to ensure an isolated route from the output side to the input side. [3]

Basic switching regulators utilize inductors, capacitors, transistors, and diodes to regulate voltage. The operation of the switching regulator is based on a transistor, which switches rapidly on and off. In these basic regulators, the energy is stored in the inductor during the on phase and subsequently released into the load during the off phase. This causes fluctuations in the output voltage, necessitating the use of a capacitor to smooth it out. Diodes are used to handle the desired polarity of the input and output voltages and provide a current path to ensure that the circuit operates properly. [4]

A practical converter, on the other hand, necessitates the inclusion of a switch controller to turn on and off the transistor. The ratio of on-time compared to the whole switching period is defined as a duty cycle:

$$D = \frac{t_{ON}}{T_{sw}}, \tag{1}$$

where

D is the duty cycle,
t_{ON} is the on-time of the switch and
T_{sw} is the switching period of the converter.

The output properties of the converter may be modified by adjusting the duty cycle. Typically, the decision to open and close the transistor is related to monitoring the output voltage or inductor current via a feedback loop. For example, to regulate a constant output voltage, a feedback loop can monitor the output voltage and alter the duty cycle accordingly. [5]

Converters can be designed to work in three modes. In continuous conduction mode (CCM), during a switching cycle, the current never falls to zero. In other words, the inductor is never reset, and its flux never goes to zero throughout the switching cycle under consideration. The current is always flowing through the coil when the switch closes. [5]

Another mode is discontinuous conduction mode (DCM). Within a switching cycle, the inductor is fully reset because the current always goes back to zero. There is no current flowing through the coil when the switch closes. [5]

The third operating mode is boundary or borderline conduction mode (BCM), also called critical conduction mode (CRM). In BCM or CRM, a controller monitors the current in the inductor, and when it reaches zero, the switch is immediately closed. In other words, to reactivate the switch, the controller always waits for the inductor current to reset. The switching period, T_{SW}, may vary if the peak current is high, and the off-slope is relatively flat. Therefore, the BCM or CRM converter is a variable-frequency system. [5]

The change of the conduction mode is not uncommon during the operation of the converter. Even if the converter is meant to work in DCM at nominal load currents, the circuit might operate in CCM at start-up until the output voltage reaches its desired level. On the other hand, a converter intended to operate in CCM at nominal load current will enter DCM as the load current decreases. These operational mode changes should not cause problems for a well-designed converter. [5]

The key benefit of a switching regulator is its high efficiency, as ideally, the diode is the only component that dissipates power since there is a finite voltage drop across it. Ideal capacitors and inductors, on the other hand, do not dissipate power. Also, an ideal transistor is a switch, which does not suffer from a voltage drop when the switch is closed and does not have current flowing through it when the switch is open. As a result, there is no power dissipation in the ideal switch since the voltage or current is always zero. This utilization of reactive components with switching leads to high efficiency, even though all components dissipate some power in the real world. The efficiency of the switching regulators typically ranges from 70 % to 95 %. [4][6][7]

High efficiency is desirable in converters since all power loss is converted into heat, which directly affects the cooling requirements of the system. Therefore, the basis for achieving larger output powers is to increase efficiency. This also brings up another advantage of switching regulators: their compact size. By minimizing the power losses, the converter design can be denser and use fewer cooling elements. [6]

Switching regulators are also versatile. They can produce an output voltage with the opposite polarity of the input voltage and can be utilized for both step-up and step-down operations. Usually, any input voltage can be converted into any required output voltage. Moreover, additional outputs can also be created by adding more secondary windings. The desired output voltage is then determined by the turn ratio of the secondary windings. [6]

Noise and electromagnetic interference (EMI) emissions, which result from the rapid switching, are the main drawbacks of the switching regulators. Also, the design is typically more complex and, therefore, more costly compared to the design of a linear regulator. [4]

In DC-DC power supplies, a number of popular converter topologies are used, each with unique benefits and drawbacks. While some topologies have fewer components and are smaller in size, others do better in high-power applications. The aim of this literature research is to determine the suitable converter topology for a low-power, small-sized battery emulator power supply. Therefore, the topologies that could perform well in this regard are described more in detail.

2.1 Buck, Boost, and Buck-Boost

A buck, a boost, and a buck-boost are the most basic switch-mode converters, from which several other converter topologies are derived. The purpose of the buck converter is to decrease the output voltage below the input voltage, whereas the boost converter is intended to increase the output voltage higher than the input voltage. The buck-boost converter is a combination of the two, which can either increase or decrease the output voltage with respect to the input voltage. All three converters utilize a switch, a diode, an inductor, and a capacitor, but the configuration is slightly different. [6]

The fundamental operating principle of the buck converter is that the output voltage equals the input voltage when the switch is closed, and the output voltage equals zero when the switch is open. As a result, the average output voltage may be adjusted between the input voltage and zero voltage by varying the duty cycle of the switch. However, a practical buck converter requires additional components, as can be seen in Figure 2. The inductor L_1 and the capacitor C_1 form a low-pass filter to reduce unwanted ripple content, and the diode D_1 operates as a freewheeling diode to provide a return path for the inductor current when the switch Q_1 is open. [6][8]

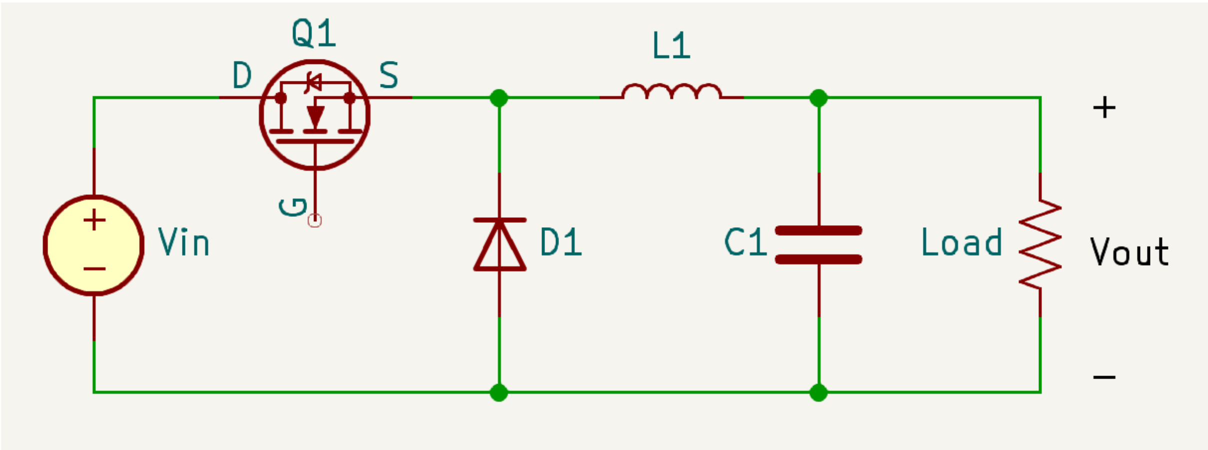


Figure 2. Schematic diagram of the buck converter.

The components of the boost converter are the same as those of the buck converter, but their arrangement is different, as can be seen in Figure 3. During the on-time, when the switch Q_1 is closed, the inductor L_1 stores energy, and the output capacitor C_1 feeds the load while the diode D_1 is reverse biased. During the off-time, when the switch Q_1 is opened, the diode D_1 becomes forward biased, and the energy of the inductor L_1 appears in series with the input source. This results in the inductor L_1 and the input source starting to supply the output together, and the current now flows into the output capacitor C_1 and the load. [5]

The buck-boost converter is a combination of the first two, although the output voltage is inverted. As shown in Figure 4, the orientation of the diode D_1 is opposite, indicating a negative output voltage. During the on-time, the operation of the buck-boost is similar to that of the boost converter. The switch Q_1 is closed, and the current in the inductor L_1 increases while diode D_1 is reverse biased. At the same time, the output capacitor C_1 supplies the load. When the switch Q_1 opens, the inductor voltage reverses, and the diode D_1 starts to conduct. The current now flows into capacitor C_1 and the load. The duty cycle of the switch defines the output voltage, which can be higher or lower in magnitude compared to the input voltage. When the duty cycle approaches one, the output voltage approaches higher in magnitude, and when the duty cycle approaches zero, the output voltage approaches lower in magnitude. [5]

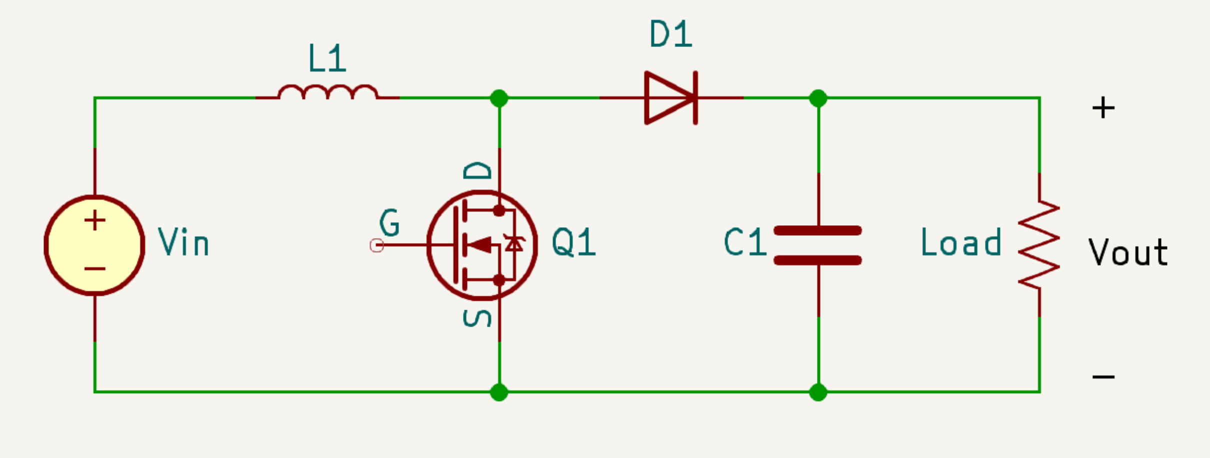


Figure 3. Schematic diagram of the boost converter.

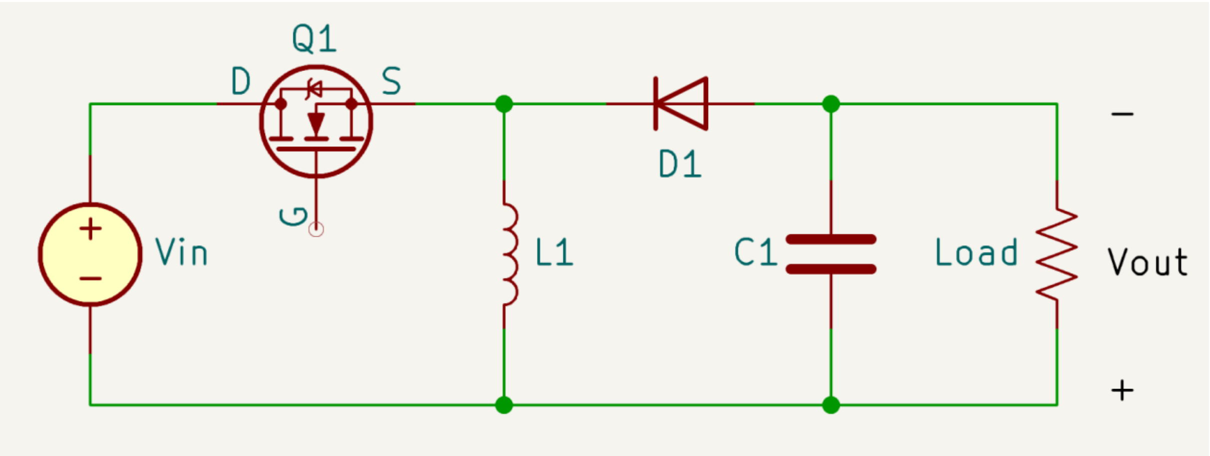


Figure 4. Schematic diagram of the buck-boost converter.

2.2 Flyback

A flyback converter is an isolated version of a buck-boost converter. It is typically used for applications requiring low power and/or high voltage in the 20 to 200 W power range. The converter is utilized in a variety of applications, including cell phone chargers, computer power supplies, xenon flash lamps, light-emitting diode (LED) lighting, lasers, and copiers. [9][10]

The schematic of the flyback converter is shown in Figure 5. The converter operates in two states. During the energy-storing state, while the switch Q_1 is turned on, the transformer T_1 stores the energy from the input source. Also, the diode D_1 is reverse biased, and no current flows to the secondary. When the switch is turned off, the power delivery state begins, and the voltage in the transformer reverses. As a result, the output diode D_1 is forward biased, and energy in the magnetizing inductance of the transformer is transferred to the output. The output polarity is defined by the transformer polarity. An inverting converter is achieved by using a noninverting transformer. [10]

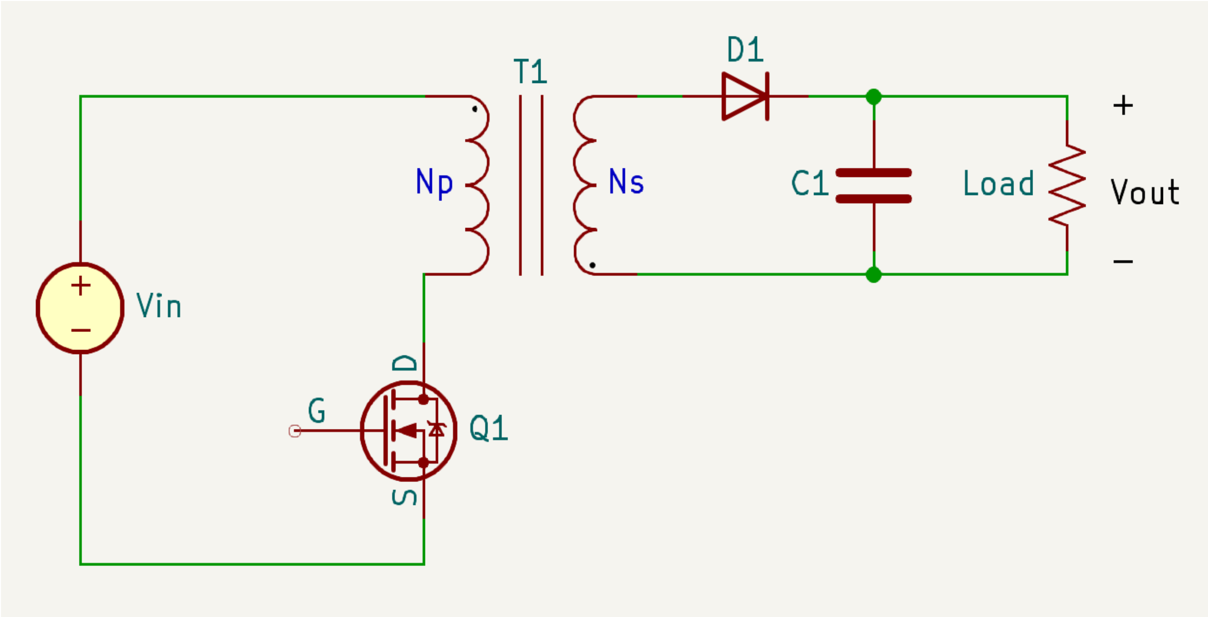


Figure 5. Schematic diagram of the flyback converter.

The flyback converter is inexpensive since it has a small total number of components. It does not require an additional output inductor since the transformer stores the energy. The flyback converter also has the capability of good transient response and voltage tracking since there are no inductors at the output stage. On the other hand, the drawback of the flyback converter is the large amount of DC current that must be stored in the transformer. As a result, a relatively large transformer is required. Therefore, the flyback suits best for applications requiring high output voltage and low output current. [4][9][10]

Flyback can be designed to operate either in CCM or DCM. The modes differ greatly in terms of component stress, output voltage control, transient responsiveness, and efficiency. While the delivered output powers are equal, the CCM causes the same voltage stress on components as the DCM, but the current stress on the components is lower in the CCM. In terms of output voltage regulation, both CCM and DCM flyback converters function equally well. However, the former shows a slower output voltage response. When it comes to efficiency, the CCM outperforms the DCM, whose conduction and switching losses are higher. On the other hand, DCM enables the use of a smaller transformer. [11]

Operating mode can be decided with the value of magnetizing inductance L_{MAG} once the switching frequency, the duty cycle, the load resistance, and the transformer turns ratio have been specified. The boundary value for the L_{MAG} is given as

$$L_{M,crit} = \frac{(1-D)^2 R_L}{2F_{sw}} \left(\frac{N_p}{N_s}\right)^2, \tag{2}$$

where

$L_{M,crit}$ is the boundary value of the magnetizing inductance,
D is the duty cycle of the switch,
F_{SW} is the switching frequency,
R_L is the load resistance,

N_p is the number of primary turns, and
N_s is the number of secondary turns.

If the L_{MAG} of the transformer is larger than the $L_{M,crit}$, the converter will operate in the CCM, and if the L_{MAG} of the transformer is smaller than the $L_{M,crit}$, the converter will operate in the DCM. [11]

The input-to-output voltage relationship of the flyback converter in CCM mode is stated as

$$V_{OUT} = V_{IN} * \left(\frac{N_s}{N_p}\right) * \left(\frac{D}{1-D}\right), \tag{3}$$

where
V_{OUT} is the output voltage,
V_{IN} is the input voltage,
N_s is the number of secondary turns,
N_p is the number of primary turns and
D is the duty cycle of the converter.

In DCM mode, the input-to-output voltage relationship of the flyback converter is expressed as

$$V_{OUT} = V_{IN} * \frac{D}{\sqrt{\frac{2*L_S}{R_L*T_{SW}}}}, \tag{4}$$

where
V_{OUT} is the output voltage,
V_{IN} is the input voltage,
D is the duty cycle of the converter,
R_L is the load resistance,
T_{SW} is the switching period and
L_S is the secondary inductance. [12]

2.3 Active Clamp Flyback

In the flyback topology, transformer leakage inductance causes high voltage and current stress for the switch when it opens. One way to handle this issue is to use a passive clamp circuit, which includes a resistor, a capacitor, and a diode across the transformer primary. The idea of the passive clamp circuit is to dissipate the energy stored in the leakage inductance and, therefore, reduce the stress on the switch. However, the energy is wasted, which makes the use of the passive clamp inefficient. [12][13]

An active clamp technique is utilized to make the clamping more efficient. As presented in Figure 6, the active clamp flyback uses the auxiliary switch Q_2 and the capacitor C_2 as a clamp circuit across the transformer primary. Similar to the passive clamp, the stress of the main switch Q_1 is reduced, but instead of dissipating the leakage inductance energy of the transformer, it is stored in the capacitor C_2. Recycling the energy later in the switching cycle also increases efficiency. The lumped capacitance C_r illustrates the sum of different capacitive elements, such as parasitic capacitances of the switches, transformer interwinding capacitance, and the capacitance reflected from the output diode. These capacitances contribute to achieving zero voltage switching (ZVS) condition for the switches, which reduce their voltage stress. [12][13][14]

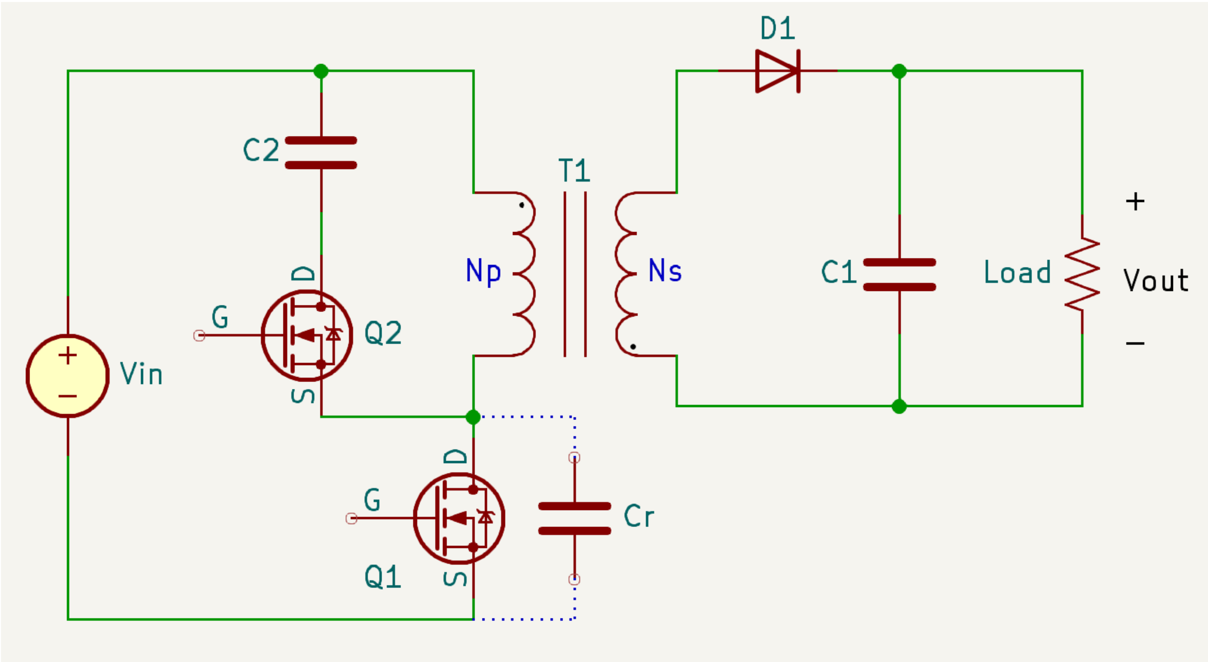


Figure 6. Schematic diagram of the active clamp flyback converter.

The entire operation of the active clamp flyback converter can be divided into eight different states [14]. However, the operation can be simplified to four intervals as can be seen in Figure 7. The steady state key waveforms in the figure are the gate-source voltage of the switches $V_{GS}(Q_1)$ and $V_{GS}(Q_2)$, the voltage across the lumped capacitance $V(C_r)$, the transformer magnetizing current $I_{MAG}(T_1)$ and the voltage across the transformer primary $V_P(T_1)$.

Similar to the standard flyback, during the first interval, t1-t2, when the switch Q_1 is conducting, the voltage across the transformer primary $V_P(T_1)$ rises and the transformer magnetizing current $I_{MAG}(T_1)$ starts to increase. As a result, the transformer T_1 stores energy while the diode D_1 on the secondary side is reverse biased. Hence, no power is delivered from the primary side to the secondary side. [5][14][15]

The second interval, t2-t3, starts when the main switch Q_1 is opened, and the primary voltage $V_P(T_1)$ starts to fall. As the primary current of the transformer is still flowing and neither of the switches is conducting, the current starts to decrease while charging the lumped capacitance C_r at the node between the switches Q_1 and Q_2 and the transformer T_1. From the moment the drain-to-source voltage of the main switch Q_1 rises over the clamp capacitor C_2 voltage, the body-diode of the auxiliary switch Q_2 begins to conduct. As a result, the auxiliary switch Q_2 can now be turned on under ZVS. [5][13][14]

The third interval, t3-t4, begins when the auxiliary switch Q_2 is closed and the diode D_1 starts to conduct. The leakage inductance is now resonating with the clamp capacitor C_2, and as a result, it creates a sinusoidal resonant current. The current flowing on the secondary side is the difference between this primary resonant current and the magnetizing current. When the resonant current finally drops to zero, it reverses and begins to flow in the opposite direction. Now the switch Q_2 can be opened. [5][14][15]

As the auxiliary switch Q_2 is opened, the fourth interval, t4-t1, begins. Similar to the second interval, neither of the switches is conducting. However, the resonant current is now reversed, leading to the secondary current starting to drop towards zero. The reversed current on the primary side now flows through the body-diode of the main switch Q_1 and discharges the lumped capacitance. This enables the ZVS condition for the main switch Q1, and as soon as the switch is opened, the next switching period starts. [5][14][15]

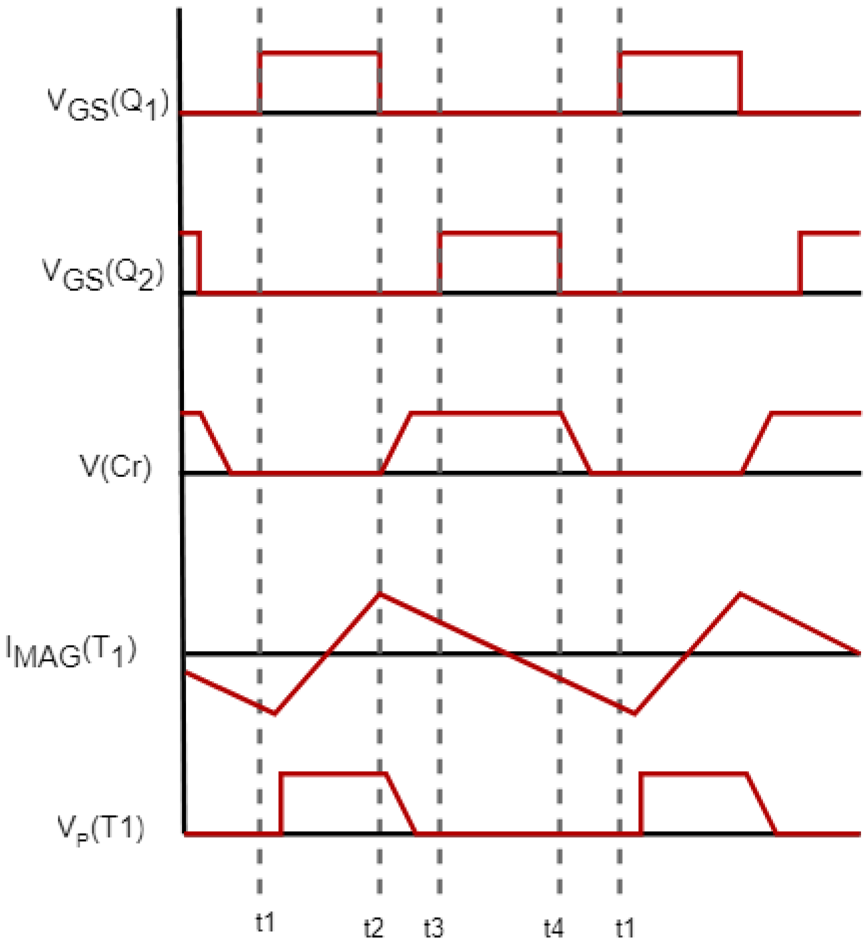


Figure 7. Steady state timing diagram of the active clamp flyback.

2.4 Single-Switch Forward

A forward converter is an isolated version of the buck converter. The transformer of the converter is used to step down or step up the primary voltage and provide galvanic isolation between the input voltage source and the output voltage.

The schematic of the single-switch forward converter is shown in Figure 8. When the switch Q_1 is closed, the transformer magnetizing current starts to build up linearly from zero to a final value with a slope of

$$I_{MAG} = \frac{V_{IN}}{L_{MAG}} \left[\frac{A}{s}\right], \tag{5}$$

where

I_{MAG} is the magnetizing current,
V_{IN} is the input voltage and

L_{MAG} is the magnetizing inductance of the transformer.

Energy goes directly through the transformer as the primary and secondary sides conduct simultaneously. The transformer dot arrangement defines that a positive voltage occurs on the secondary side, forward biasing the diode D_1. At the same time, diodes D_2 and D_3 become reverse biased. The sum of the magnetizing current I_{MAG} and the inductor current I_L reflected on the primary side equals the total current flowing through the primary side and switch Q_1 during the on-time. [5][6][15]

When the switch Q_1 opens, the transformer magnetizing current must continue to flow. To avoid saturation of the transformer and destroying the switch Q_1, it is critical to reset the magnetizing current during the switch off-time. This is done by adding a reset winding N_r with an opposite polarity compared to the primary and secondary windings. Negative voltage will be induced at the dot end of the reset winding, which causes the diode D_3 to conduct and allows the magnetizing current to reset. As soon as the magnetizing current is fully reset, the voltage across the transformer stays zero until the next switching cycle begins. [16]

During the switch off-time, also D_2 becomes forward biased, which completes the current path for the output. In the meantime, D_1 is reverse biased. Equation (6) shows the relationship between the input and output voltages of the forward converter in the CCM operation:

$$V_{OUT} = V_{IN} * \left(\frac{N_s}{N_p}\right) * D, \tag{6}$$

where

V_{OUT} is the output voltage,
V_{IN} is the input voltage,
N_s is the number of secondary turns,
N_p is the number of primary turns and
D is the duty cycle of the converter. [6]

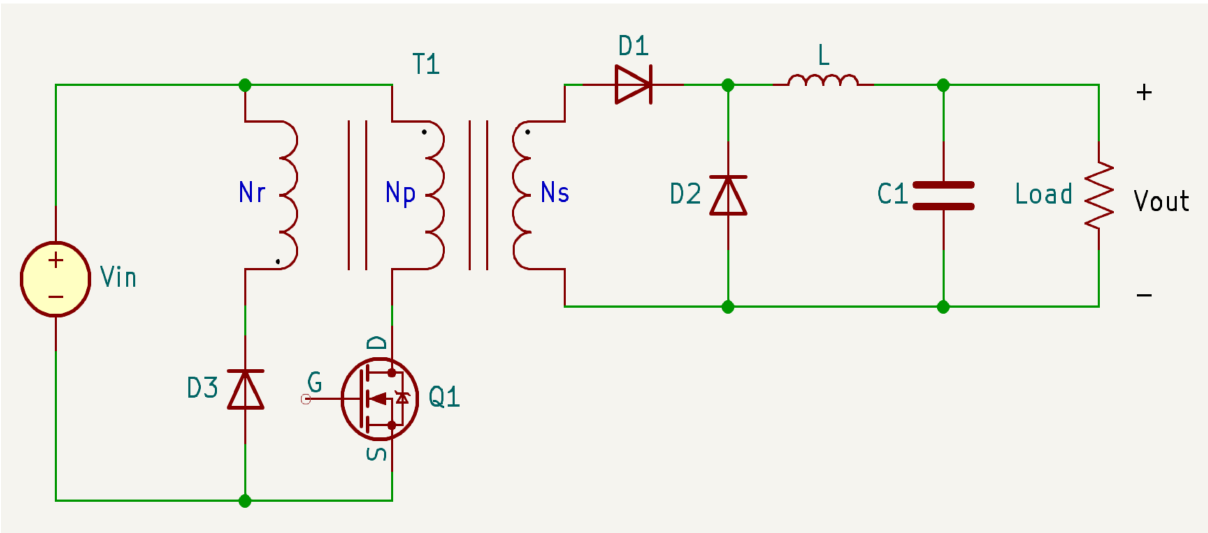


Figure 8. Schematic diagram of the single-switch forward converter.

The maximum duty cycle of the single-switch forward is 50 % in the condition when the number of reset winding turns N_r equals the number of primary winding turns N_p. However, the maximum duty cycle can be greater than 50% if the Nr is lower than the Np, but as a tradeoff, the voltage stress of the switch will increase. This also applies in the other direction; if the N_r is greater than the N_p, the voltage stress of the switch will be lower, but the maximum duty cycle will also be lower. [16]

The transformer core is not employed as energy storage in the forward topology, which distinguishes it from the flyback topology. Therefore, the transformer does not require an air gap, and it can have a much higher magnetizing inductance. This results in lower peak currents and, therefore, lower conduction losses. Also, the output inductor and freewheeling diode of the forward converter keep the output current relatively constant; as a result, the secondary ripple current is drastically reduced compared to the flyback. [9][17]

The forward converter also has its disadvantages compared to the flyback. The cost of the forward converter is higher since it requires an extra output inductor and a freewheeling diode. Furthermore, forward converters are intended to function at maximum power in the CCM and only enter the DCM in light-load situations. This results in a minimum load requirement since gain varies drastically when the converter enters the DCM. Also, the switch must be capable of handling a high voltage, which is around two times the input voltage in the single-switch forward converter. [16][17]

The forward converter is most suitable for low-power applications, just like the flyback. Typically, when a high output current is required, the forward topology is preferable over the flyback topology, and when the requirement is a high output voltage, it goes the other way around.

2.5 Two-Switch Forward

A two-switch forward converter, as the name implies, is a forward converter with two switches. By adding a second switch, the voltage stress of the switches can be reduced by half. This allows the two-switch forward to operate at higher power levels than the single-switch forward.

The schematic of the two-switch forward converter is shown in Figure 9. As it is seen in the schematic, the difference between the single-switch forward and the two-switch forward is the additional switch with a diode, and there is no need for the reset winding. Two identical gate drive signals control the switches Q_1 and Q_2. During on-time, when the switches are closed, the transformer T_1 primary is given the input voltage V_{IN}, and the total current flowing through the switches and the transformer primary is the sum of the magnetizing current and the reflected output inductor current. At the same time, the transformer transfers energy from the primary side to the secondary side. [16][18]

When the switches Q_1 and Q_2 are turned open, the voltage across the transformer primary winding reverses. This forward biases diodes D_3 and D_4, allowing the magnetizing current to flow back to the source V_{IN}, resetting the transformer core. The voltage across the primary winding and the switches is equal to the input voltage if there is no leakage inductance in the transformer. Therefore, the voltage across the switches is half the value compared to the single-switch forward. After the magnetizing current is fully reset, the diodes D_1 and D_2 become reverse biased, and the next cycle can begin. The secondary side of the two-switch forward operates similarly compared to the single-switch forward. Also, the output-input voltage relationship of the two-switch forward is equal to the single-switch forward, which was introduced in equation (6). [16][18]

The transformer is fully reset if the off-time is longer than the on-time. As a result, the maximum duty cycle of the two-switch forward converter cannot exceed 50 %. Furthermore, the peak current sets the upper limit for the power range. An excessively high peak current would make managing the losses across the switches impractical and necessitate a large transformer. Therefore, the two-switch forward converter is usually used at power levels lower than 350 W. [16][18]

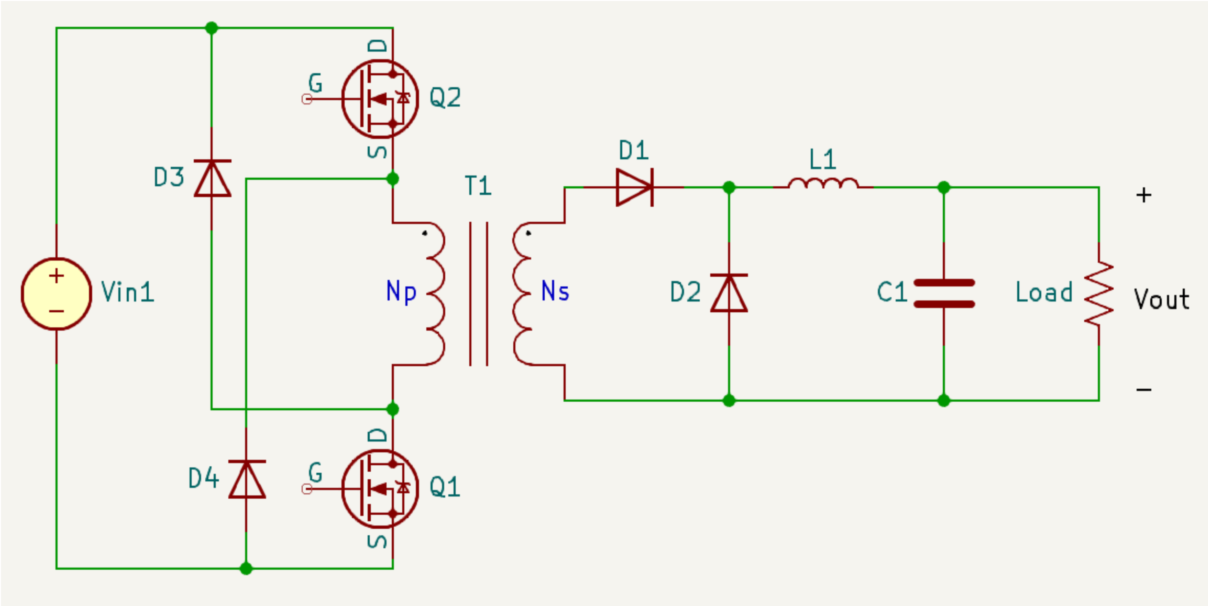


Figure 9. Schematic diagram of the two-switch forward converter.

2.6 Active Clamp Forward

Active clamp forward topology employs a similar clamp circuit as the active clamp flyback to absorb and recycle the parasitic energy while resetting the transformer core. It allows ZVS, and compared to other forwards, it offers lower EMI, a wider duty cycle range, and lower switch voltage stress. [19]

On the other hand, the active clamp forward also has its disadvantages. Without an accurate duty clamp, utilizing the wider duty cycle range can lead to transformer saturation or unnecessary voltage stress on the switch. Also, the delay time between the two switches on the primary side must be synchronized. [19]

Similar to the two-switch forward architecture, there are two switches, but they are configured differently. The second switch, called an auxiliary switch, is for utilizing the clamp capacitor. The secondary side of the converter is exactly the same as in the other forward topologies.

The active clamp forward converters come in two varieties: a low-side clamp and a high-side clamp. Differences between the two are minor but still significant since both produce a different transfer function and, therefore, a different clamp voltage during the reset. This affects the choice of components in terms of gate drive circuitry and the voltage rating of the clamp capacitor.

Applying the clamp circuit across the drain-to-source of the main Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET) is known as the low-side clamp, as seen in the schematic of the low-side active clamp forward in Figure 10. The switch Q_1 operates as an N-channel main switch, and the switch Q_2 is a P-channel auxiliary switch. The clamp capacitor C_1 is placed between the drains of the switches, resulting in the drain-to-source voltage of the main switch Q_1 equaling the clamp voltage. For the low-side clamp, the transfer function between the input voltage and the clamp voltage is defined as

$$V_{DS(Q1)} = V_{CL} = \left(\frac{1}{1-D}\right) * V_{IN}, \tag{7}$$

where

$V_{DS(Q1)}$ is the drain-to-source voltage of the main switch Q_1,
V_{CL} is the clamp voltage,
D is the duty cycle of the converter and
V_{IN} is the input voltage. [20]

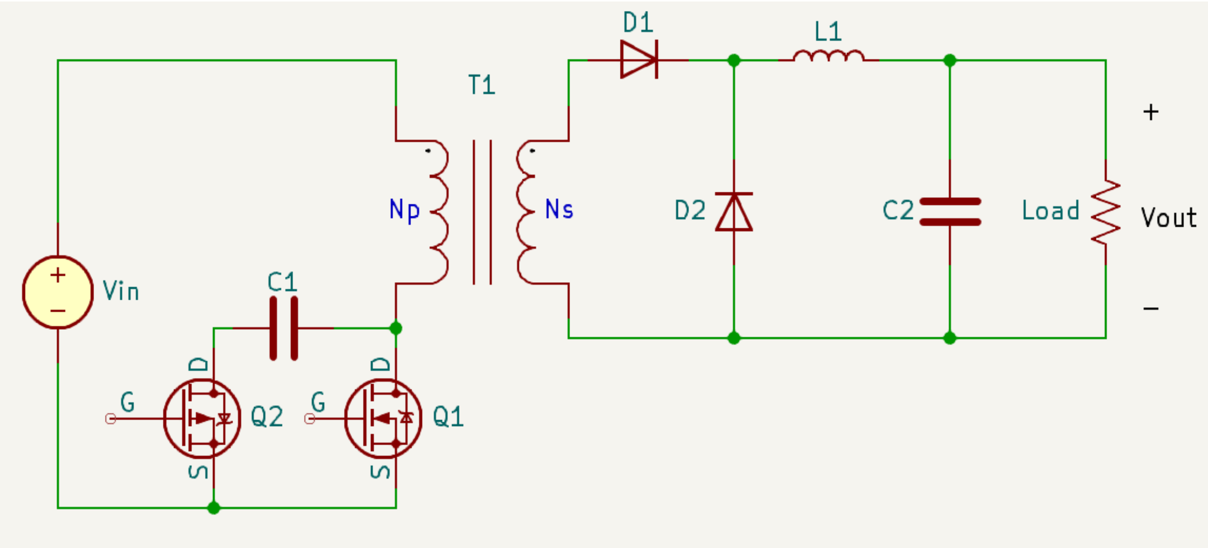


Figure 10. Schematic diagram of the low-side active clamp forward.

The entire operation of the low-side active clamp forward can be divided into eleven different states [21]. However, a basic and simplified operation can be explained with four switching intervals, as seen in the steady-state timing diagram in Figure 11. The diagram shows the gate-source voltage of the switches $V_{GS}(Q_1)$ and $V_{GS}(Q_2)$, the drain-source voltage of the main switch, which equals the clamp capacitor voltage $V_{DS/CL}(Q_1/C_1)$, the transformer magnetizing current $I_{MAG}(T_1)$, the voltage across the transformer primary $V_P(T_1)$ and the output voltage V(out) during the different intervals.

During the first interval t_0-t_1, the main switch Q_1 is closed under ZVS, while Q_2 is open. The primary side current, which is the sum of the transformer magnetizing current and the reflected secondary current, builds up linearly. On the secondary side, diode D_1 conducts, while diode D_2 is reverse biased. During this state, energy is transferred from the primary to the secondary, and the state ends when the main switch Q_1 is opened. [22][23]

During the interval t_1-t_2, the main switch Q_1 is opened under ZVS, resulting in the primary current continuing to flow through the body diode of the auxiliary switch Q_2. For this reason, a low-side active clamp necessitates the use of the P-channel auxiliary switch. On the secondary side, the diode D_1 is now reverse biased, and the diode D_2 is now conducting. As a result, there is no reflected primary current. Since only the transformer magnetizing current flows through the body-diode of the switch Q_2, conduction loss is negligible, and the switch may activate under ZVS. The resonant period, which is the time between turning off the main switch and turning on the auxiliary switch, is what separates the active clamp method from other singleended transformer reset techniques. At the end of the interval, the magnetizing current has reached its maximum positive value, and the converter is ready for the next interval. [22]

The third interval t_2-t_3 begins when the auxiliary switch Q_2 starts to conduct. Since the magnetizing current has already peaked at the maximum positive value, the primary side current begins to flow smoothly in the opposite direction, resulting in the transformer primary being reset. The voltage difference between the clamp voltage and the input voltage is now seen across the transformer primary, as shown in equation (8):

$$V_{RESET} = V_{CL} - V_{IN}, \tag{8}$$

where

V_{RESET} is the voltage in primary during the transformer reset,
V_{CL} is the clamp voltage and
V_{IN} is the input voltage.

On the secondary side, diode D_1 is reverse biased, and diode D_2 is forward biased, just like in the previous time frame. [22]

The fourth and last interval of the switching cycle t_3-t_4 is also a resonant period. At the start of the interval, the auxiliary switch Q_2 is opened under ZVS, resulting in the primary current continuing to flow in the opposite direction through the body diode of the main switch Q_1. As the body diode of the main switch Q_1 conducts, the switch has a condition to close under ZVS. However, in certain low-load situations, there might be some turn-on losses. Gradually, the primary current starts to change its direction again since the maximum negative peak value has already been reached. On the secondary side, the current is still circulating through the diode D2, while the diode D_1 is reverse biased until the next switching cycle begins after the interval t_3-t_4 is completed. [22]

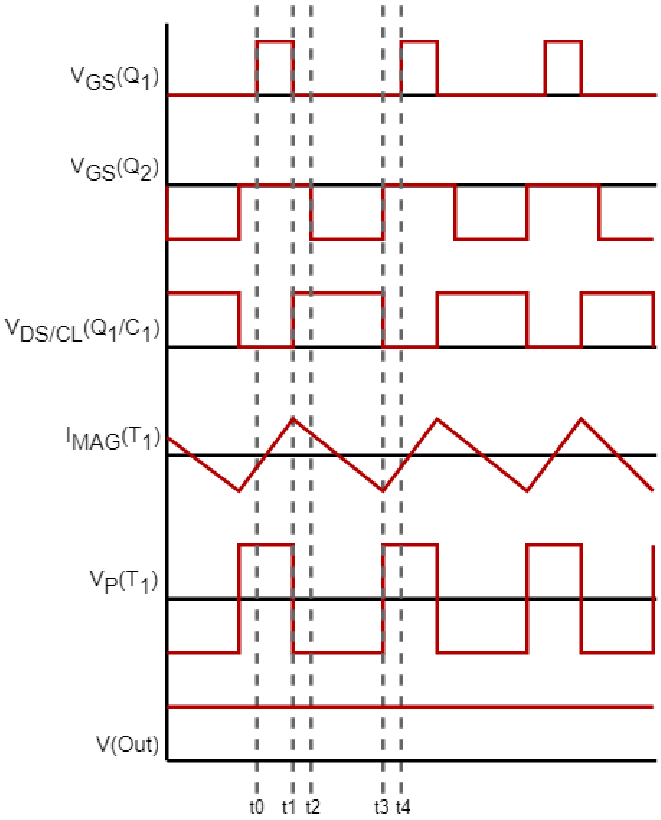


Figure 11. Steady state timing diagram of the low-side active clamp forward.

The power transfer of the high-side clamp is similar to the low-side clamp; when the main switch is closed, the input voltage is applied across the transformer magnetizing inductance, and power is delivered to the secondary side. However, resetting the transformer differs from the low-side clamp. The high-side clamp requires using an N-channel device as an auxiliary switch, for the same reason as the low-side clamp needs a P-channel device, direction of the body diode. Also, an additional transformer is needed to drive the gate if the controller is missing a built-in high-side driver. [20]

As seen in Figure 12, the high-side clamp has the clamp circuit on the high-side, across the transformer primary. When the auxiliary switch Q_2 is closed, the clamp voltage is applied across the transformer magnetizing inductance. Compared to the low-side scenario, where the clamp voltage V_{CL} is applied across the drain-to-source junction of the primary switch, this is significantly different. For the high-side clamp, the transfer function between the input voltage and the clamp voltage is defined as

$$V_{RESET} = V_{CL} = \left(\frac{D}{1-D}\right) * V_{IN}, \tag{9}$$

where

V_{CL} is the clamp voltage,
D is the duty cycle of the converter and
V_{IN} is the input voltage.

The clamp voltage of the high-side clamp configuration also equals the voltage in the primary during the transformer reset. [20]

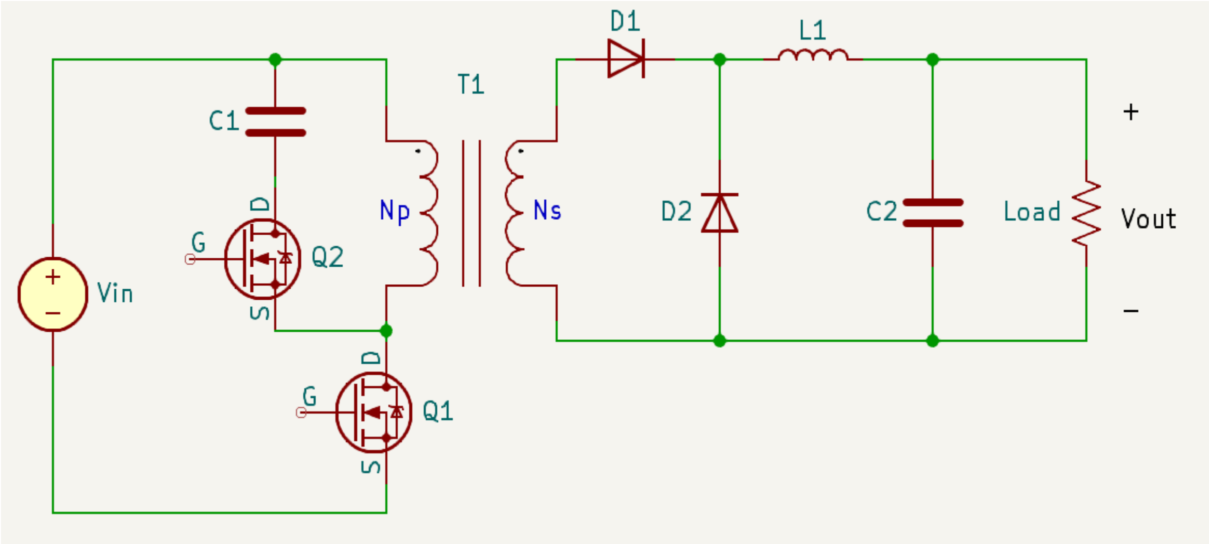


Figure 12. Schematic diagram of the high-side active clamp forward.

In terms of timing, the main difference between the high-side and the low-side clamps is the gate drive voltage of the auxiliary switch. As the N-channel switch is required, the gate drive voltage of the switch must also be asynchronously out of phase with the main switch Q_1, as seen in Figure 13.

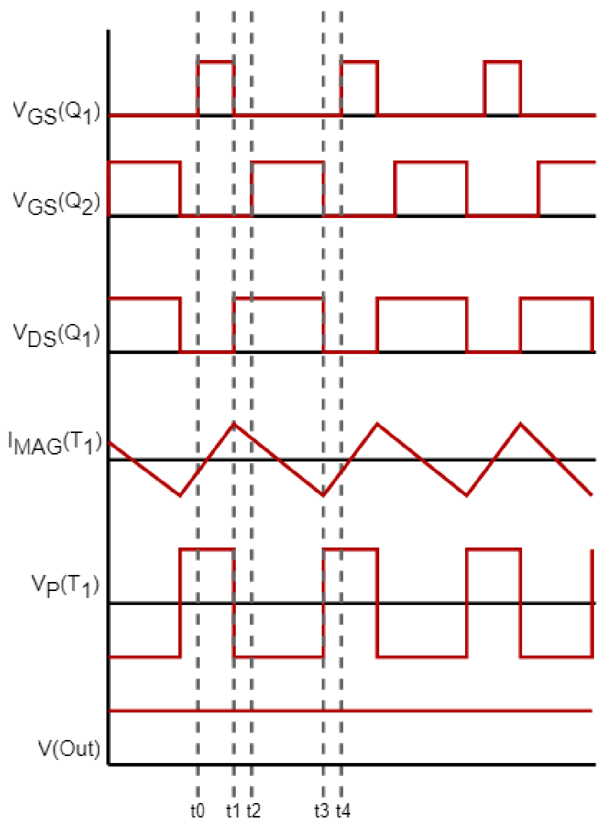


Figure 13. Steady state timing diagram of the high-side active clamp forward.

The drain-to-source voltage of the main switch Q_1 is identical to the low-side clamp, which was defined earlier in equation (7). Additionally, voltage across the primary winding is the same in both configurations during the transformer reset. However, the difference comes in the clamp voltage. Comparing the transfer functions reveals that the low-side clamp has a clamp voltage that is greater by V_{IN} volts:

$$\Delta V_{CL} = V_{CL(LS)} - V_{CL(HS)} \tag{10}$$

$$\Delta V_{CL} = \left(\frac{1}{1-D}\right) * V_{IN} - \left(\frac{D}{1-D}\right) * V_{IN} = V_{IN}, \tag{11}$$

where

$V_{CL(LS)}$ is the low-side clamp voltage and
$V_{CL(HS)}$ is the high-side clamp voltage.

In conclusion, employing the high-side clamp results in less voltage stress. However, at minimal input voltage and maximum duty cycle, the voltage tends to rise more swiftly. Hence, the maximum duty cycle must be very carefully limited to avoid going above the maximum drain-to-source voltage of the main switch. Additionally, the high-side clamp circuit may necessitate an extra gate drive transformer, raising the cost of the design. On the other hand, the low-side clamp eliminates the need for the gate drive transformer and allows better control over the clamp voltage despite its larger value. Also, the maximum duty cycle limitation is not so critical since it does not significantly affect the clamp voltage fluctuation. [20]

3 POWER SUPPLY UNIT AND CONVERTER DESIGN

The IPPSuM 1.0 power supply unit has been designed for battery-powered device testing. Its maximum supply voltage of 10 V and maximum output power of 30 W are too low for modern standards. For example, USB Type-C with USB PD 3.0 supports voltages up to 20 V [24]. By expanding the output voltage range from 0 - 10 V to 0 - 20 V and the maximum output power from 30 W to 100 W, the number of possible devices that can be tested increases significantly.

A block diagram of the original IPPSuM 1.0 power supply unit is shown in Figure 14. The red parts on the block diagram illustrate the power transfer of the power supply. As can be seen, there are two identical output channels that can function independently, offering the capability to both source and sink current. Each channel can operate at distinct voltage levels and even in different modes simultaneously. This allows, for example, testing a device on one channel by providing power while simultaneously testing its charger on the other channel by sinking current.

The power supply operates with the nominal input voltage of 12 V, which is first isolated using an active clamp forward DC-DC converter. Depending on the load and the output voltage setting, the output of the DC-DC converter was measured to be between 5.18 V and 11.45 V. The converter is then followed by the linear regulator, which basically acts like a variable resistor to regulate the voltage to the range of 0 to 10 V. Finally, the regulated voltage is supplied to the output of the power supply.

The power supply monitors the output with a feedback circuit, shown as a red dashed line in the figure. The output is fed back from the linear regulator to the controller of the DC-DC converter through operational amplifiers and an optocoupler. The yellow parts in the figure illustrates the current measurement function, which is achieved using shunt resistor, while the blue parts show the control and data paths. Basically, the power supply is controlled by a computer which acts as a master controller. The power supply works as a slave by utilizing a 10/100 Mbps Local Area Network (LAN) connection to receive commands from the master and send the readings back. In other words, the user sets the output voltage and current, as well as reads them using a microcontroller (MCU).

The focus of this thesis is the DC-DC switch-mode converter since it is necessary to be modified in order to raise the overall output voltage and power ranges of the power supply. To fulfill the output voltage requirement of the new power supply, the maximum output voltage of the DC-DC converter must be increased. Figure 15 shows the target output voltages for the new IPPSuM 2.0, with the blue box being the focus of this thesis. Since the output voltage of the converter was measured from the old design to be approximately 1.5 V higher than the output voltage of the power supply, the maximum output voltage of the converter should be targeted at close to 21.5 V. The minimum output voltage should stay between 5 V and 6 V, as it was in the old design, and therefore, it is targeted to be 5.5 V. By closely meeting the target voltages, the linear regulator should be able to fine-tune the output voltage to the final value.

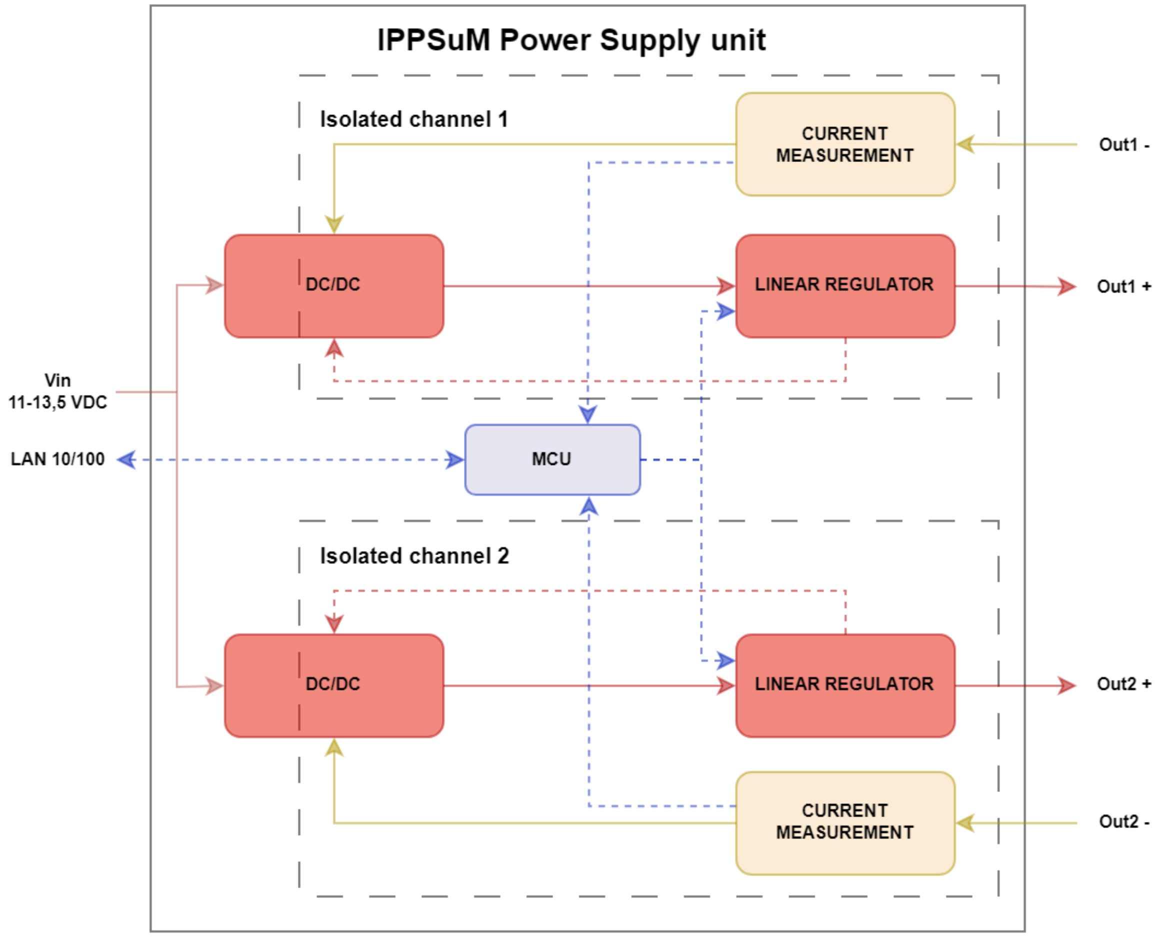


Figure 14. Block diagram of the IPPSuM 1.0 power supply.

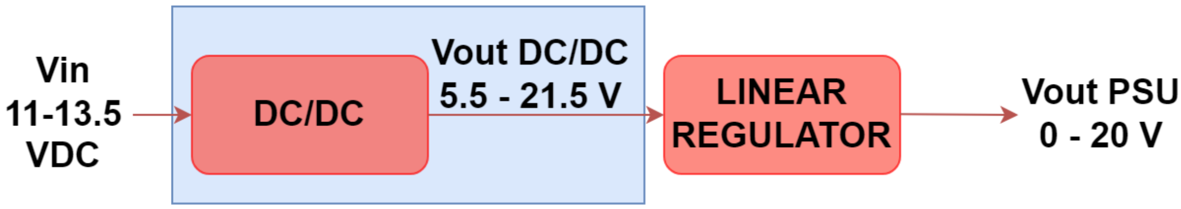


Figure 15. Output voltage requirements of the new IPPSuM 2.0.

Except for output voltage and output power, all other specifications are in line with the previous design while maintaining the physical dimensions of the IPPSuM 1.0 is also desirable. A list of specifications for the IPPSuM 2.0's DC-DC converter may be found in Table 1. In order to ensure clarity, inputs and outputs are now seen from the perspective of the DC-DC converter.

Table 1. Specification of the DC-DC converter of the IPPSuM 2.0

Parameter	Condition	Min	Max	Unit
Input voltage		11	13.5	VDC
Output voltage		5 - 6	21-22	VDC
Output current		0	5	A
Output power		0	107.5	W
Switching frequency			300	kHz
Duty cycle			0.7	
Overshoot / Undershoot	Dynamic load 0.2 A – 1.5 A. Slew rate 0.5 A / µs		100	mV

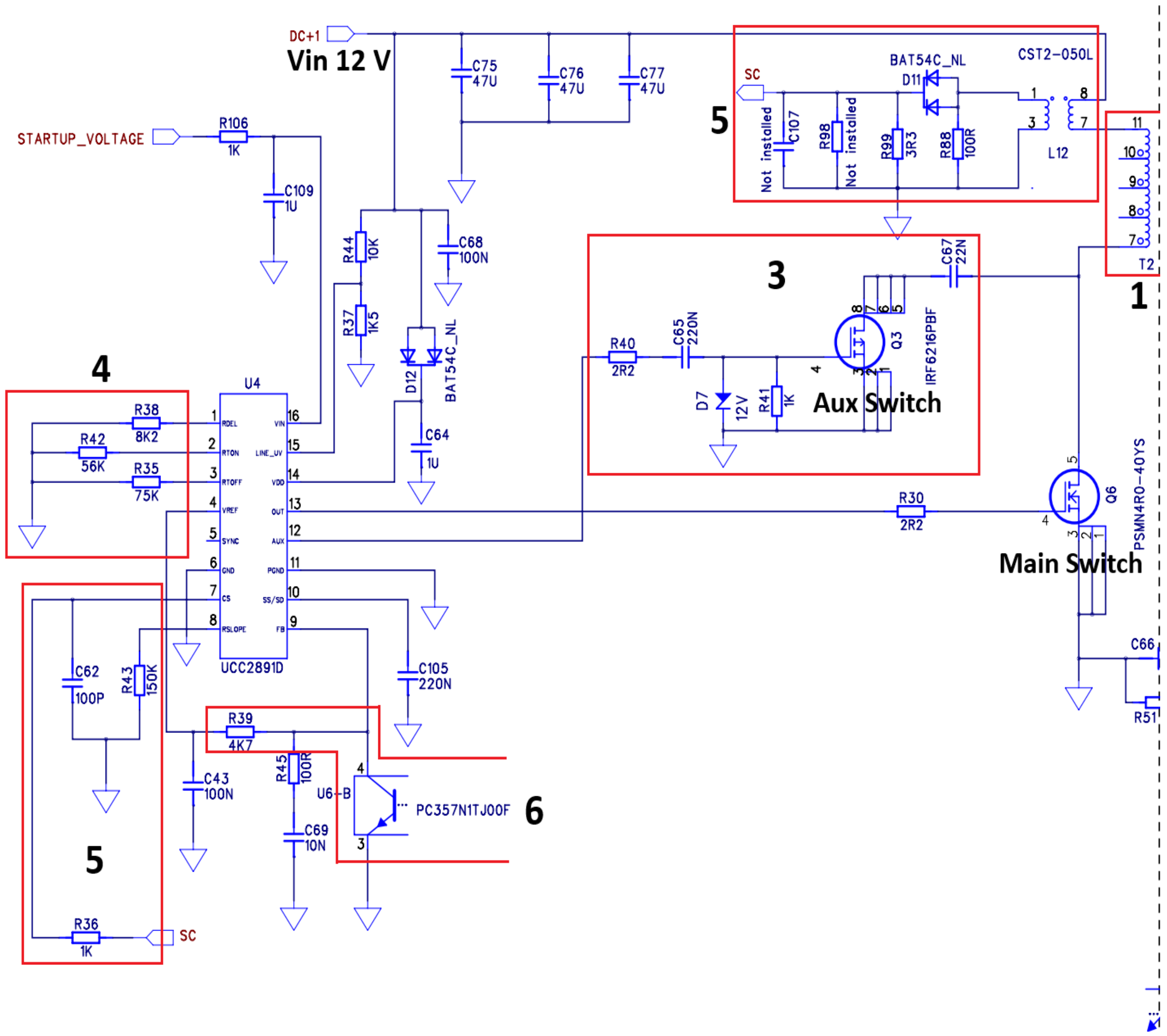
3.1 Implementation

The first step is to choose the most suitable topology and a controller for it. A wide range of duty cycle that active clamp forward topology can achieve, is necessary in order to achieve a compact sized converter with a wide output voltage range. Since the IPPSuM 1.0 also employed the active clamp forward topology, using and optimizing the original design is an ideal way to upgrade the converter. The previous design's active clamp Pulse-Width Modulation (PWM) controller UCC2891 from Texas Instruments may be utilized, since it is still manufactured, and the only significant modifications are to the output voltage range and maximum output power. Therefore, the converter design is based on the old IPPSuM 1.0 design and the datasheet of the PWM controller UCC2891 [25].

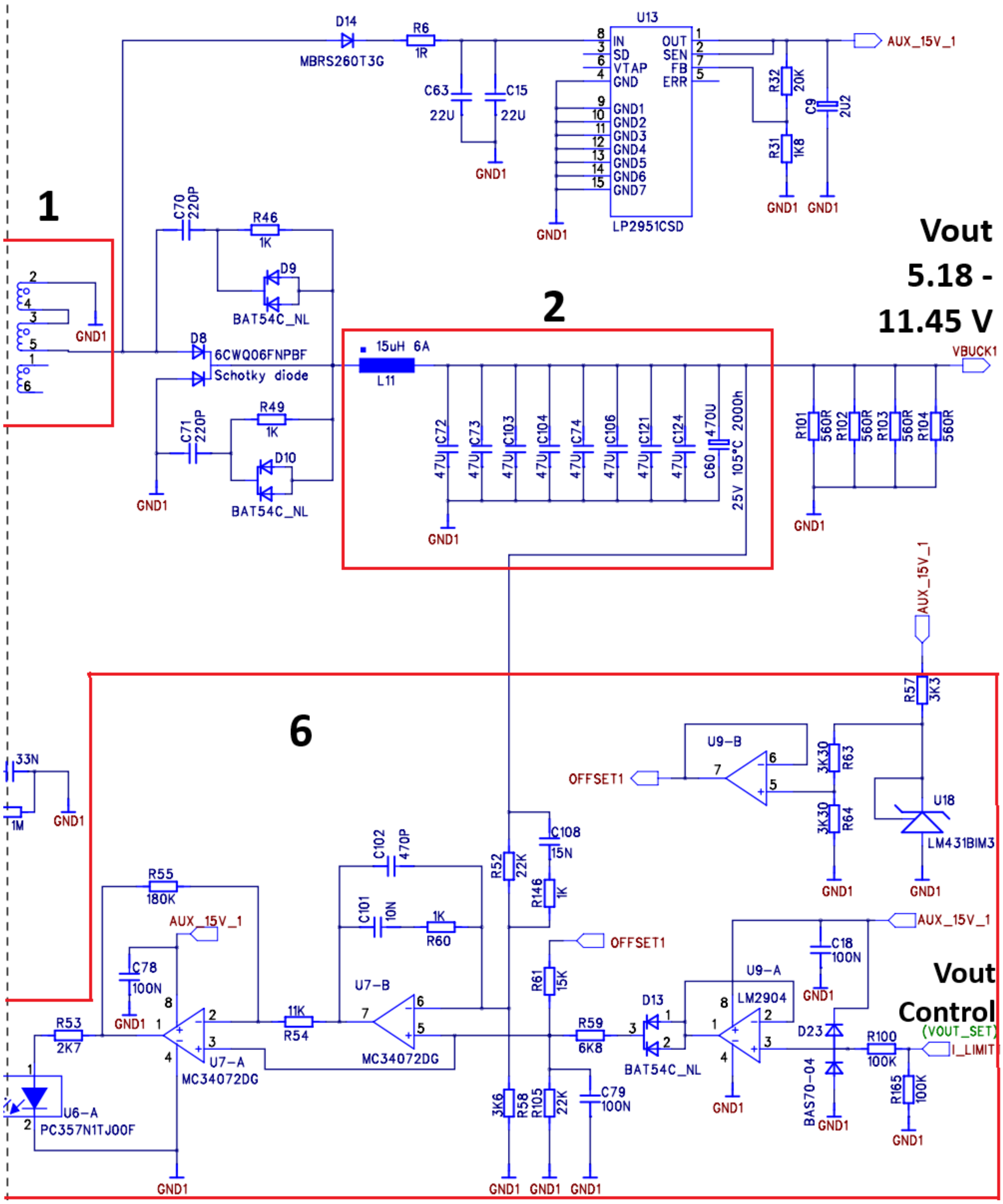
Figure 16 shows the DC-DC switch-mode converter of the old IPPSuM 1.0. The figure is separated into two sections: the primary side (a) and secondary side (b) of the converter. Certain areas of the converter have to be resized according to the new specification, and therefore the parts and components are marked and numbered to correspond with the design and sizing order of these parts in the thesis. The parts are as follows: the power transformer (1), output inductor and capacitors (2), active clamp circuit (3), resistors that determine the duty cycle and switching frequency (4), current sensing circuitry (5), and feedback loop (6).

Briefly explained, the parts of the converter are as follows: On the primary side (a), DC+1 is the input voltage of 12 V, which is supplied by the external AC/DC power. The input voltage switching across the power transformer T2 (1) is done by the main switch Q6, causing energy to transfer to the secondary side. Between the power transformer T2 and the input source is the current sensing transformer L12 as a part of the current sensing circuit (5), which monitors the input current drawn. The active clamp circuit (3) lowers the voltage stress of the main switch Q6 and increases the efficiency of the converter by recycling the parasitic energy of the power transformer. The UCC2891D on the left side of the figure (a) is the PWM controller, which controls the switching of the switches Q6 and Q3. Their switching frequency and duty cycle are determined by the resistors R38, R42, and R35 (4).

On the secondary side (b), the transferred energy passes through the diode rectifier to the output inductor and capacitors (2), where the energy is stored and filtered. On the right side of the schematic is the output of the DC-DC converter, which the linear regulator sees as an input voltage. The converter supplies the output voltage of 5.18 to 11.45 V depending on the voltage setting and load current. The output voltage is set to the desired value using the feedback loop (6). The feedback loop compares the output voltage to the reference voltage coming from the MCU, and if there is difference between the voltages, an error signal is generated. It is then fed via the optocoupler U6 to the UCC2891D controller on the primary side of the converter. The duty cycle of the switches is adjusted according to this error signal.



(a) Primary side of the DC-DC converter of the IPPSuM 1.0.



(b) Secondary side of the DC-DC converter of the IPPSuM 1.0.

Figure 16. The DC-DC switch-mode converter of the IPPSuM 1.0. Marked and numbered parts have to be resized.

3.2 Transformer

The power transformer is sized first, and it corresponds to the part (1) in Figure 16. The purpose of the transformer is to transfer energy from the primary to the secondary side of the converter. The winding ratio of the transformer determines whether the voltage increases or decreases on the secondary side. The winding ratio can be estimated using equation (12) since we know that the minimum input voltage is 11 V, the target output voltage is 21.5 V, and the maximum duty cycle of the UCC2891 controller is 70 %.

$$\frac{N_P}{N_S} = \frac{V_{IN(MIN)}}{V_{OUT(MAX)}} * D_{MAX} = \frac{11\ V}{21.5\ V} * 0.7 \approx 0.36, \tag{12}$$

where

N_P is the number of turns on primary side,
N_S is the number of turns on secondary side,
$V_{IN(MIN)}$ is the minimum input voltage,
$V_{OUT(MAX)}$ is the maximum output voltage and
D_{MAX} is the maximum duty cycle of the converter.

As the value is difficult to translate to the number of turns, the result is rounded to the value of 0.33. Therefore, the secondary side will have three turns for every one turn in the primary. [22]

It is desired that the most critical components of the design are realistic and have Simulation Program with Integrated Circuit Emphasis (SPICE) models made by the manufacturer. Therefore, Würth's 750341181 planar transformer was found suitable for the design [26]. The datasheet states that the primary side magnetizing inductance is 86 µH; however, since the transformer must be used in the opposite direction, equation (13) must be used to get the secondary side magnetizing inductance.

$$L_{MAG(S)} = \left(\frac{N_S}{N_P}\right)^2 * L_{MAG(P)} = \left(\frac{1}{0.33}\right)^2 * 86\ \mu\mathrm{H} \approx 789.7\ \mu H, \tag{13}$$

where

$L_{MAG(S)}$ is the magnetizing inductance of the secondary side and
$L_{MAG(P)}$ is the magnetizing inductance of the primary side. [6]

The minimum duty cycle of the converter can be calculated with equation (12) but using the lowest output voltage and maximum input voltage as follows in equation (14).

$$D_{MIN} = \frac{N_P}{N_S} * \frac{V_{OUT(MIN)}}{V_{IN(MAX)}} = \frac{2}{6} * \frac{5.5\ V}{13.5\ V} \approx 0.14, \tag{14}$$

where

D_{MIN} is the minimum duty cycle,
$V_{OUT(MIN)}$ is the minimum output voltage and
$V_{IN(MAX)}$ is the maximum input voltage. [22]

3.3 Output Inductor

The output inductor filters the output current while it is also used as an energy storage [17]. In Figure 16, it can be found belonging to the part (2). A larger inductor inductance value results in reduced output ripple current flowing through the inductor, but as a trade-off, it usually requires more space on the printed circuit board (PCB). Conversely, though, the root-mean-square (RMS) ripple current entering the output capacitors increases when more ripple current passes the inductor. The ripple current value is not specified, but it is selected to be 30 % of the maximum output current, since it is considered as a good trade-off value for most applications [27]. The output inductor value is calculated using equations (15) and (16).

$$L_O = \left(\frac{V_{OUT(MAX)}}{\Delta I_{LO(\%)} * I_{OUT(MAX)} * F_{sw}}\right) * (1 - D_{MIN}) \tag{15}$$

$$L_O = \left(\frac{21.5\ V}{0.3*5\ A*300\ kHz}\right) * (1 - 0.14) \approx 41\ \mu H, \tag{16}$$

where

L_O is the inductance of the inductor,
$V_{OUT(MAX)}$ is the maximum output voltage,
$\Delta I_{LO(\%)}$ is the maximum allowed ripple current in percents,
$I_{OUT(MAX)}$ is the maximum output current,
F_{SW} is the switching frequency of the converter and
D_{MIN} is the minimum duty cycle of the converter.

According to the calculation, the most optimal output inductor value for the new design is 41µH. In order to define the minimum current rating of the output inductor, the ripple current is first back-calculated using the maximum voltage that the power supply should be able to deliver. Using equations (17) and (18), the maximum ripple current is:

$$\Delta I_{LO} = \left(\frac{V_{OUT}}{L_O * F_{sw}}\right) * (1 - D_{MIN}) \tag{17}$$

$$\Delta I_{LO} = \left(\frac{21.5\ V}{41\ \mu H * 300\ kHz}\right) * (1 - 0.14) \approx 1.75\ A_{pp} \tag{18}$$

Using the highest output voltage results in an estimated maximum ripple current of 1.75 A_{pp}. The RMS inductor current $I_{LO(RMS)}$ can then be calculated using the maximum ripple current and the maximum output current of the power supply in equations (19) and (20).

$$I_{LO(RMS)} = \sqrt{I_{OUT(MAX)}^2 + \frac{\Delta I_{LO}^2}{3}} \tag{19}$$

$$I_{LO(RMS)} = \sqrt{(5\ A)^2 + \frac{(1.75\ A_{pp})^2}{3}} \approx 5.10\ A_{RMS}, \tag{20}$$

where

$I_{LO(RMS)}$ is the RMS inductor current and
ΔI_{LO} is the maximum allowed ripple current.

The peak current seen by the secondary can be calculated as follows

$$I_{LO(PK)} = I_{OUT(MAX)} + \frac{\Delta I_{LO}}{2} = 5\ A + \frac{1.75\ A_{PP}}{2} \approx 5.88\ A_{PK}, \tag{21}$$

where

$I_{LO(PK)}$ is the secondary peak current. [22]

3.4 Output Capacitors

Together with the output inductor, the output capacitors are also grouped to part (2) in Figure 16. The primary goal of the output capacitors is to reduce output voltage ripple [17]. They also affect the transient response of the converter, which means the time it takes for the output voltage to reach a steady state during load changes. Since transient response must be considered in the design, the transient overshoot characteristics of the old design can be used to calculate the minimum output capacitance. The voltage overshoot and undershoot must be limited to 0.1 V as the load current is stepped between 0.2 A and 1.5 A. As shown in equations (22) and (23), the output capacitance can be calculated by equating the inductive energy with the capacitive energy.

$$C_O = \frac{L_O * \left(I^2_{STEP(MAX)} - I^2_{STEP(MIN)}\right)}{\left(V^2_{OS(MAX)} - V^2_{OS(MIN)}\right)} \tag{22}$$

$$C_O = \frac{41\ \mu H * ((1.5\ A)^2 - (0.2\ A)^2)}{((21.6\ V)^2 - (21.5\ V)^2)} \approx 21\ \mu F, \tag{23}$$

where

C_O is the output capacitance,
L_O is the output inductance,
$I_{STEP(MAX)}$ is the maximum of the step load,
$I_{STEP(MIN)}$ is the minimum of the step load,
$V_{OS(MAX)}$ is the absolute maximum output voltage with the overshoot and
$V_{OS(MIN)}$ is the maximum stabile output voltage.

The value of the minimum output capacitance impacts only the capacitive component of the output ripple. When filtering the switching frequency ripple, the equivalent series resistance (ESR) of the filter capacitor usually matters more than its actual capacitance [28]. Therefore, the maximum R_{ESR} of the output capacitor is calculated in equations (24) and (25).

$$R_{ESR(MAX)} = \frac{\Delta V_{OUT(RIP)}}{\Delta I_{LO}} \tag{24}$$

$$R_{ESR(MAX)} = \frac{30\ mV}{1.75\ A_{pp}} \approx 17\ m\Omega, \tag{25}$$

where

$R_{ESR(MAX)}$ is the maximum ESR value of the capacitor,
$\Delta V_{OUT(RIP)}$ is the maximum allowed ripple voltage and
ΔI_{LO} is the maximum allowed ripple current.

The output capacitors have to be selected to satisfy the minimum capacitance of 21 μF while having a maximum ESR of 17 mΩ and a minimum ripple current rating of 5.10 A_{RMS}. A quicker transient response can be achieved by using a smaller output capacitance. Since the output capacitance value is proportional to the output inductor value, a smaller capacitance can be used by lowering the inductor value, for example, by using a higher switching frequency or allowing more ripple current through the inductor. [22]

3.5 Active Clamp Circuit

The active clamp circuit is marked as part (3) in Figure 16. The main purpose of the active clamp circuit is to lower the voltage stress of the main switch and increase the efficiency of the converter by recycling the parasitic energy of the power transformer [23]. This in turn allows using higher duty cycle. As a low-side clamp configuration is used, an auxiliary switch necessitates using a P-channel device because of the direction of the body diode. A low gate charge of the switch is more important than a low $R_{DS(ON)}$ since only the transformer magnetizing current, which is fairly low, is flowing through the switch.

Given that the auxiliary switch needs to be a ground-referenced P-channel device and demands a negative gate drive voltage, a gate drive circuit is required to drive it since output voltages below ground reference are not produced by the UCC2891 controller. An example of the gate drive circuit can be seen in Figure 17. As the UCC2891's AUX output first rises to a positive value, the capacitor C_{AUX} is charged, but since the Schottky diode D_{AUX} is forward biased, the voltage at the gate of the auxiliary switch Q_{AUX} is near zero. As soon as the AUX output is switched off and falls to zero volts, the voltage at the gate of Q_{AUX} decreases by the same amount, well below zero volts. Since the diode D_{AUX} is now reverse biased and the resistor R_{AUX} is large enough, the capacitor C_{AUX} cannot discharge instantaneously. This results in a negative gate drive voltage for the auxiliary switch, Q_{AUX}.

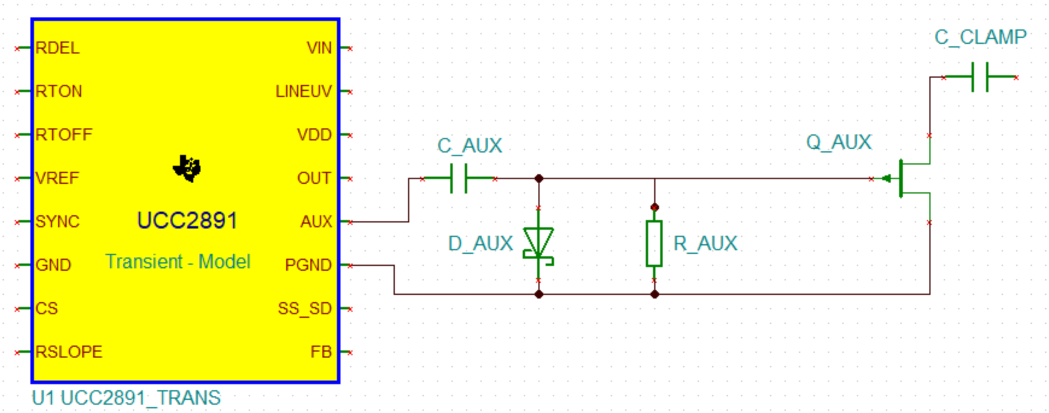


Figure 17. Low-side gate drive circuit example of the auxiliary switch. [22]

In order to provide a negative driving voltage for the P-channel switch, it is crucial that the voltage across C_{AUX} stays fairly constant and the maximum peak value is zero volts. This is accomplished by sizing R_{AUX} and C_{AUX} so that the time constant is significantly larger than the PWM period, as seen in equation (26):

$$R_{AUX} * C_{AUX} \approx \frac{100}{F_{SW}}, \tag{26}$$

where

R_{AUX} is the resistance of the R_{AUX} resistor,
C_{AUX} is the capacitance of the C_{AUX} capacitor and
F_{SW} is the switching frequency.

By choosing R_{AUX} arbitrarily to be 1 kΩ, C_{AUX} can be solved as seen in equation (27):

$$C_{AUX} = \frac{100}{1\ k\Omega * 300\ \text{kHz}} = 0.33\ \mu\text{F} \tag{27}$$

As the auxiliary switch is conducting during the transformer reset period, the voltage on the transformer primary is determined by the difference between the input and the clamp voltage. The clamp voltage can be calculated using equations (7) and (28) to define the minimum voltage ratings for the clamp capacitor and the switches. [22]

$$V_{DS(QMAIN)} = V_{CL} = \left(\frac{1}{1-0.7}\right) * 13.5\ V = 45\ V \tag{28}$$

The value of the clamp capacitor, C_CLAMP, in figure 17, must be sufficiently high to act as a continuous voltage source. It is mostly determined by the maximum allowable ripple voltage, which may be sustained. A greater capacitance value decreases voltage ripple but makes the transient response slower. On the other hand, a faster transient response is achieved using smaller capacitance, but at the price of increased voltage ripple. In equations (29) and (30), the clamp capacitor value, C_{CLAMP}, is approximated straightforwardly, ensuring that the transient performance is not compromised from the perspective of the active clamp circuit. The value is calculated so that the resonant time constant is significantly larger than the maximum off-time:

$$C_{CLAMP} > 10 * \left(\frac{(1-D_{MIN})^2}{L_{MAG} * (2 * \pi * F_{SW})^2}\right) \tag{29}$$

$$C_{CLAMP} > 10 * \left(\frac{(1-0.14)^2}{789.7\ \mu H * (2 * \pi * 300\ kHz)^2}\right) \approx\ 2.6\ nF, \tag{30}$$

where

C_{CLAMP} is the capacitance of the clamp capacitor,
D_{MIN} is the minimum duty cycle,
L_{MAG} is the magnetizing inductance of the transformer and
F_{SW} is the switching frequency.

As 2.6 nF is a good starting point for the clamp capacitor, the value might need tuning after the ripple voltage is measured in a real circuit. [22]

3.6 Maximum Duty Cycle and Switching Frequency

The maximum duty cycle D_{MAX} and the switching period T_{SW} of the circuit are determined by the three resistors: R_{DEL}, R_{TON}, and R_{TOFF}. In Figure 16 (a), part (4), the resistors correspond to

R38, R42, and R35, respectively. The UCC2891 controller use pin number 12 (AUX) and 13 (OUT) as gate drive signal outputs to drive the main and auxiliary switches, which can be seen in Figure 16. The resistor R_{DEL} is used to set an identical turn-on delay for both the signals. The resistor R_{TON} sets the charge current of the internal timing capacitor, while the resistor R_{TOFF} sets the discharge current of the internal timing capacitor. The charging and discharging delays of the capacitor determine the switching period T_{SW} and the maximum duty cycle D_{MAX} of the circuit. [28]

The value of the R_{DEL} is decided to be the same as in the old design, which was 8.2 kΩ. Using the value, the delay times t_{DEL1} and t_{DEL2} can be calculated in order to size the other resistor values. The delay times t_{DEL1} and t_{DEL2} are calculated as follows:

$$t_{DEL1} = t_{DEL2} = 11.1 * 10^{-12} * R_{DEL} + 15 * 10^{-9}\ [s] \quad (31)$$

$$t_{DEL1} = t_{DEL2} = 11.1 * 10^{-12} * 8.2\ \text{k}\Omega + 15 * 10^{-9} \approx\ 1.06 * 10^{-7}\ s, \quad (32)$$

where

t_{DEL1} is the delay time between AUX to OUT,
t_{DEL2} is the delay time between OUT to AUX and
R_{DEL} is the resistance of the R_{DEL} resistor.

As specified earlier, the maximum duty cycle has to be set to 70 %, while the switching frequency is 300 kHz. According to these specifications, the values of the resistors R_{TON} and R_{TOFF} can be calculated. The on-time t_{ON} is first calculated using equations (33) and (34):

$$D_{MAX} = \frac{t_{ON}}{T_{SW}} \quad (33)$$

$$t_{ON} = D_{MAX} * T_{SW} = 0.7 * \frac{1}{300\ kHz} \approx 2.33 * 10^{-6}\ s, \quad (34)$$

where

D_{MAX} is the maximum duty cycle,
t_{ON} is the on-time and
T_{SW} is the switching period.

Using the on-time t_{ON}, the value of the resistor R_{TON} is then calculated as follows:

$$t_{ON} = 36.1 * 10^{-12} * R_{TON} - t_{DEL1}\ [s] \quad (35)$$

$$\text{R}_{\text{TON}} = \frac{t_{ON} - t_{DEL1}}{36.1*10^{-12}} = \frac{2.33*10^{-6}\ s - 1.06*10^{-7}\ s}{36.1*10^{-12}} \approx 61.6\ k\Omega, \quad (36)$$

where

R_{TON} is the resistance of the resistor.

Knowing the switching period T_{SW} and the on-time t_{ON}, the off-time t_{OFF} can be calculated:

$$T_{SW} = t_{ON} + t_{OFF} \quad (37)$$

$$t_{OFF} = T_{SW} - t_{ON} = \frac{1}{300\ kHz} - 2.33 * 10^{-6} \approx 1.00 * 10^{-6}\ s, \tag{38}$$

where

t_{OFF} is the off-time.

Finally, the value of the R_{TOFF} resistor can be calculated by knowing the off-time t_{OFF}, in equations (39) and (40):

$$t_{OFF} = 15 * 10^{-12} * R_{TOFF} + t_{DEL1} + 170 * 10^{-9}\ [s] \tag{39}$$

$$R_{TOFF} = \frac{t_{OFF} - t_{DEL1} - 170 * 10^{-9}}{15 * 10^{-12}} = \frac{1.00 * 10^{-6}\ s - 1.06 * 10^{-7}\ s - 170 * 10^{-9}}{15 * 10^{-12}}$$

$$\approx 48.3\ k\Omega, \tag{40}$$

where

R_{TOFF} is the resistance of the resistor.

In conclusion, the resistors R_{DEL}, R_{TON}, and R_{TOFF} must have values of 8.2 kΩ, 61.6 kΩ, and 48.3 kΩ, respectively, in order for the circuit to function at the maximum duty cycle of 70% and at the switching frequency of 300 kHz.

3.7 Feedback Loop and Current Sensing

The duty cycle is adjusted via a feedback loop so that it responds appropriately to the demands of the load. The UCC2891 controller uses current mode control, meaning it senses the current from the primary side and the output voltage regulation from the secondary side to modulate the on-time of the main switch of the converter.

The basic operation of the feedback loop, which can be seen as part (6) in Figure 16, is as follows: the output voltage on the secondary side is compared to the reference voltage, and if there is a difference between the output voltage and the reference voltage, an error signal is generated. The error signal is then transmitted to the FB-pin of the UCC2891 controller on the primary side via an optocoupler. The signal is internally buffered within the UCC2891, and its signal level is modified by a voltage divider to match the signal level of the current sensing circuit before it is connected to a PWM comparator. The current sense circuit is connected to the CS-pin of the UCC2891, which is internally connected to the other side of the PWM comparator. By comparing the voltage error signal with the current sensing circuit signal, the duty cycle is defined accordingly. [28]

The current on the primary side can be sensed with a current sense resistor or a current sense transformer. Methods differ from each other in terms of simplicity and efficiency. The sense resistor does not need other components, but it is connected in series with the main switch, meaning the losses can rise impractically high when the primary current is high. In contrast, the current sense transformer approach has a higher component count but also better efficiency since the primary current is sensed after it has been reduced by the transformer turns ratio. [22]

The IPPSuM 1.0 uses the current sense transformer method, as can be seen in Figure 16, part (5). In order to achieve the specified maximum output power of the converter of the IPPSuM 2.0, the turns ratio of the current sense transformer has to be increased from 50 to 100. Since the transformer turns ratio is changed, it is also necessary to recalculate the slope compensation. The slope compensation is determined by two resistors, R_F and R_{SLOPE} (R36 and R43 in Figure 16). R_F is the current sense filter resistor, which forms a low-pass filter together with the filter capacitor C_F (C62 in Figure 16). The value of the R_F is selected based on the corner frequency of the low-pass filter. A good starting point is to select the corner frequency to be 10 times the switching frequency, while the value of the capacitor C_F should stay between 47 pF and 270 pF. As the value of C_F is selected to be 100 pF, the value of R_F can be calculated as follows:

$$R_F = \frac{1}{2*\pi*(10*F_{SW})*C_F} = \frac{1}{2*\pi*(10*300\ kHz)*100\ pF} \approx 530\ \Omega, \qquad (41)$$

where

R_F is the current sense filter resistor,
F_{SW} is the switching frequency and
C_F is the current sense filter capacitor. [22]

Since the output inductor current reflects from the secondary to the primary, which is seen as a voltage slope across the current sense resistor R_{CS} (R99 in Figure 16), the voltage equivalent compensation ramp must be defined. In order to calculate the ramp, the value of the current sense resistor R_{CS}, as well as the primary peak current $I_{PRI(PK)}$ and primary magnetizing current I_{MAG}, must first be known. The primary magnetizing current I_{MAG} is first calculated in equation (42), followed by the calculation of the peak current $I_{PRI(PK)}$ in equation (43):

$$I_{MAG} = \frac{V_{IN(MIN)}*D_{MAX}}{F_{SW}*L_{MAG}} = \frac{11\ V*0.7}{300\ kHz*789.7\ \mu H} \approx 0.03\ A \qquad (42)$$

$$I_{PRI(PK)} = \frac{I_{LO(PK)}}{N} + I_{MAG} = \frac{5.88\ A_{PK}}{0.33} + 0.03\ A \approx 17.85\ A_{PK}, \qquad (43)$$

where

I_{MAG} is the primary magnetizing current,
$V_{IN(MIN)}$ is the minimum input voltage,
D_{MAX} is the maximum duty cycle,
F_{SW} is the switching frequency,
L_{MAG} is the power transformer magnetizing inductance,
$I_{PRI(PK)}$ is the primary peak current and
N is the power transformer turns ratio.

The UCC2891 controller has a current sense threshold of 0.75 V. Using the value of the threshold, the primary peak current, and the decided current sense turns ratio, the current sense resistor value can be determined as follows:

$$R_{CS} = \frac{V_{CS}}{I_{PRI(PK)}*\frac{1}{N_{CS}}} = \frac{0.75}{17.85\ A_{PK}*\frac{1}{100}} \approx 4.2\ \Omega, \qquad (44)$$

where

R_{CS} is the current sense resistor,
V_{CS} is the current sense threshold,

$I_{PRI(PK)}$ is the primary peak current and
N_{CS} is the current sense transformer turns ratio.

As the current sense resistor value is known, the voltage equivalent compensation ramp can be calculated:

$$\frac{dV_L}{dt} = \frac{(V_{IN(MIN)}*N_S - V_{OUT(MAX)}*N_P)*N_S*R_{CS}}{N_P^2*L_O*N_{CS}} = \frac{(11\ V*6-21.5*2)*6*4.2\Omega}{2^2*41\ \mu H*100}$$

$$\approx\ 35.3\frac{kV}{\mu s}, \tag{45}$$

where
$\frac{dV_L}{dt}$ is the voltage equivalent compensation ramp,
N_S is the secondary turns of the power transformer,
V_{OUT} is the maximum output voltage,
N_P is the primary turns of the power transformer,
L_O is the inductance of the output inductor.

Finally, the value of the resistor R_{SLOPE} can be calculated using the values of R_F and dV_L/dt. The only unknown variable is m, which determines the level of desired slope compensation. When using current mode control, the value of m should range between 0.5 and 1, hence 0.75 is selected [22]. The value of the R_{SLOPE} is calculated as follows:

$$R_{SLOPE} = \frac{5*2\ V*R_F}{\left(\frac{D_{MAX}}{F_{SW}}\right)*m*\frac{dV_L}{dt}} = \frac{5*2\ V*530\ \Omega}{\left(\frac{0.7}{300\ kHz}\right)*0.75*35.3\frac{kV}{\mu s}}$$

$$\approx\ 85.8\ k\Omega \tag{46}$$

The output inductor and capacitor of the forward converter cause a -180° phase shift and a -40 dB/decade steep gain slope. Therefore, type III compensation networks are commonly used with forward converters since they can provide over 70° phase boost and achieve a quick transient response. The compensation network necessitates six resistors, six capacitors, and an error amplifier, as shown in Figure 18. [29]

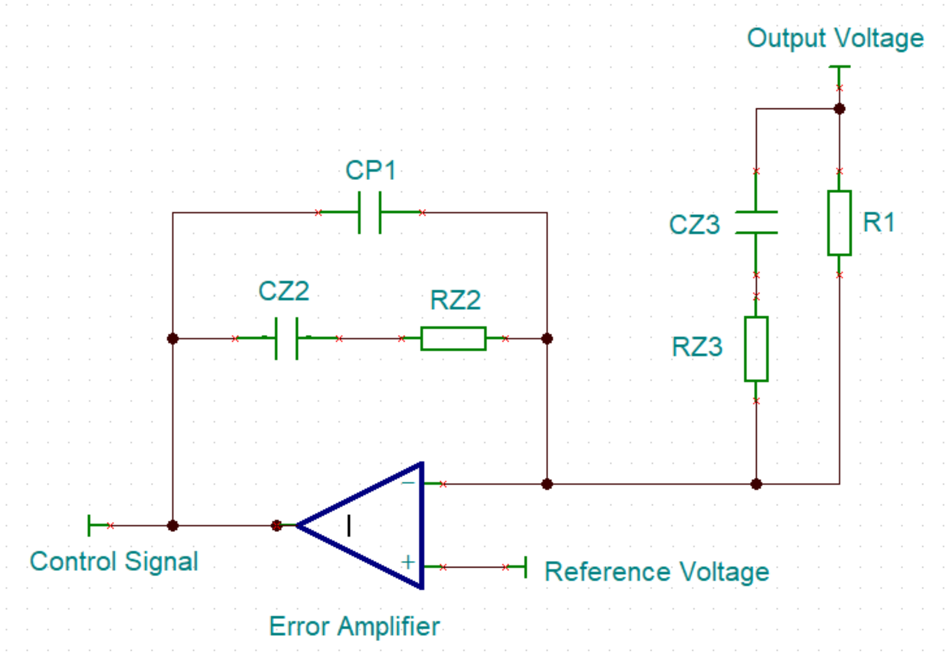


Figure 18. Type III compensation loop. [29]

The IPPSuM 1.0 has a complex feedback loop as can be seen in Figure 16, part (6). It also includes the type III compensation network, which is around the U7-B error amplifier in the figure. Therefore, the output of the U7-B in Figure 16 corresponds to the output of the error amplifier in Figure 18.

The feedback loop of the new IPPSuM 2.0 is a simplified version of the old one, employing only the type III compensation network. Accurate sizing of the compensation network's component values is necessary to avoid converter instability and regulation failure, and therefore the components are resized for the new feedback loop. When sizing the components, they must be chosen so that R1 >> RZ3 and CZ2 >> CP1. The first step is to choose the resistor R1, which is now chosen to be 30 kΩ. Placing the second zero at 60 % of the output filter's double-pole frequency, the capacitor CZ3 can be solved as follows:

$$CZ3 = \frac{1}{zsf * R1 * \frac{1}{\sqrt{LC}}} = \frac{1}{0.6 * 30\ k\Omega * \frac{1}{\sqrt{41\ \mu H * 21\ \mu F}}} \approx 1.6\ nF, \tag{47}$$

where

CZ3 is the capacitance of the capacitor,
zsf is the zero-scale factor,
R1 is the resistance of the resistor,
L is the inductance of the output inductor and
C is the capacitance of the output capacitor.

The UCC2891 controller's useful voltage range for the FB pin is between 1.25 V and 4.5 V. Therefore, the ramp amplitude is set to 3.25 V. Also, the cross-over frequency is typically set to 1/5 or 1/10 of the switching frequency, which is now chosen to be 30 kHz. The resistor RZ2 can then be calculated:

$$RZ2 = \frac{(2*\pi*fc)^2*L*C*+1}{2*\pi*fc*CZ3} * \frac{V_{RAMP}}{V_{IN}} = \frac{(2*\pi*30\ kHz)^2*41\ \mu H*21\ \mu F*+1}{2*\pi*30\ kHz*1.6\ nF} * \frac{3.25\ V}{12\ V}$$

$$\approx 28\ k\Omega, \tag{48}$$

where

RZ2 is the resistance of the resistor,
fc is the cross-over frequency,
V_{RAMP} is the amplitude of the ramp signal and
V_{IN} is the input voltage of the converter.

The capacitor CZ2 can be calculated next since the first zero is set to coincide with the second zero:

$$CZ2 = \frac{1}{zsf*RZ2*\frac{1}{\sqrt{LC}}} = \frac{1}{0.6*28\ k\Omega*\frac{1}{\sqrt{41\ \mu H*21\ \mu F}}} \approx 1.75\ nF \tag{49}$$

The capacitor CP1 can be calculated as the first pole is set to the switching frequency of the converter:

$$CP1 = \frac{1}{2*\pi*RZ2*F_{SW}} = \frac{1}{2*\pi*28\ k\Omega*300\ kHz} \approx 19\ pF, \tag{50}$$

where

F_{SW} is the switching frequency.

Finally, RZ3 can be solved by placing the second pole also at the switching frequency: [30]

$$RZ3 = \frac{1}{2*\pi*CZ3*F_{SW}} = \frac{1}{2*\pi*1.6\ nF*300\ kHz} \approx 332\ \Omega \tag{51}$$

The last step is to design the optocoupler circuit, which is shown in Figure 19. As shown in the figure, the output of the compensation loop's error amplifier (Figure 18) is connected directly to the cathode of the optocoupler's LED, while V_{OPTO} has a constant supply voltage of 15 V. The FB shown in the figure is connected to the FB pin of the UCC2891 controller, which operates within the range of 1.25 V to 4.5 V without pulse skipping. When the voltage is below 1.25 V, the controller enters pulse-skipping mode. The 5 V reference voltage V_{REF} of the UCC2891 is used for the optocoupler output, as the resistor R_{VREF} is used as a pull-up between the V_{REF} and FB pins. As the datasheet of the controller states, the maximum source current of the reference voltage is 5 mA, meaning the I_{REF} current must be limited well below this. [28]

Different optocouplers have different Current Transfer Ratios (CTR), meaning how the LED current I_{OPTO} is transferred to the current I_{REF}. Also, within a single optocoupler, the CTR value varies as the I_{OPTO} current, or temperature, varies. Additionally, as optocouplers age, their performance decreases, and this aging process accelerates as I_{OPTO} current increases [31]. Therefore, it is preferred to have the LED current, I_{OPTO}, as low as possible.

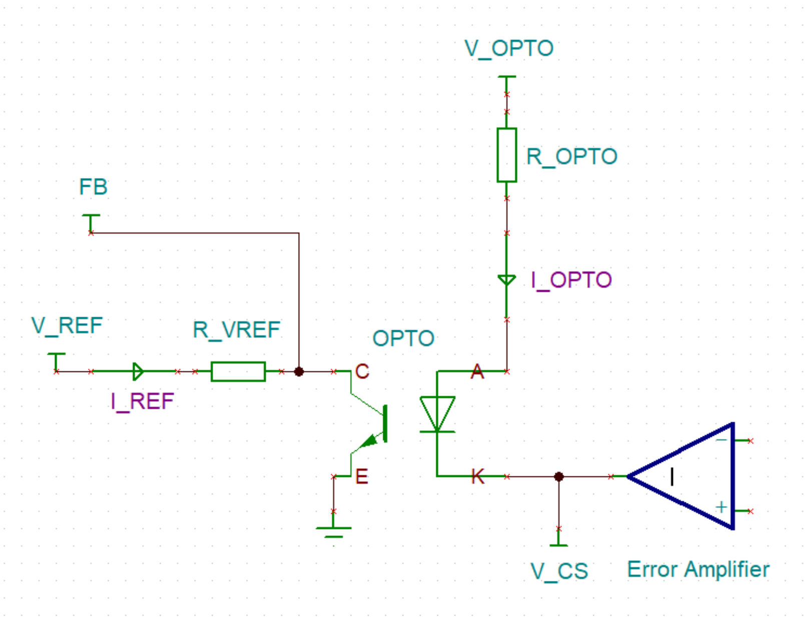


Figure 19. Optocoupler Circuit.

Assuming the CTR of the optocoupler is 200 % while the LED current I_{OPTO} is only 0.8 mA, the current I_{REF} can be calculated using equation (52):

$$I_{REF} = I_{OPTO} * CTR = 0.8\,mA * 2 = 1.6\,mA, \tag{52}$$

where

I_{REF} is the reference current,
I_{OPTO} is the LED current and
CTR is the Current Transfer Ratio of the optocoupler.

The resistor value of the R_{VREF} can then be calculated using the minimum feedback voltage as follows:

$$R_{VREF} = \frac{V_{REF} - V_{FB(MIN)}}{I_{REF}} = \frac{5\,V - 1.25\,V}{1.6\,mA} \approx 2.4\,k\Omega, \tag{53}$$

where

R_{VREF} is the resistance of the resistor,
V_{REF} is the reference voltage of the UCC2891 and
$V_{FB(MIN)}$ is the minimum feedback voltage.

As stated earlier, the voltage V_{OPTO} is connected to the 15 V supply of the error amplifier, and the voltage V_{CS} is connected to the output of the error amplifier, which varies between 0 and 15 volts. As the desired maximum I_{OPTO} current is 0.8 mA, and the forward voltage of the optocoupler is assumed to be 1.2 V. The resistor R_{OPTO} can be determined as follows:

$$R_{OPTO} = \frac{V_{OPTO} - V_F - V_{CS(MIN)}}{I_{OPTO(MAX)}} = \frac{15\text{ V} - 1.2\text{ V} - 0\text{ V}}{0.8\,mA} \approx 17.3\,k\Omega, \tag{54}$$

where

R_{OPTO} is the resistance of resistor,
V_{OPTO} is the supply voltage,
V_F is the forward voltage of the optocoupler and
$V_{CS(MIN)}$ is the minimum control signal of the error amplifier. [22]

4 SIMULATIONS

As the component sizing is done, the circuit can be built and simulated using Tina-Ti software. The software is a SPICE-based analog simulation tool developed by Texas Instruments together with DesignSoft [32]. The software was chosen since a lot of components of Texas Instruments are pre-modeled for Tina-Ti, including the most complex part of the converter in this thesis, the active clamp PWM controller UCC2891. Also, other components that have a SPICE model can be used by utilizing the built-in macro wizard.

Initially, the design was simulated using the defined component values. However, certain component values had to be optimized based on the results, as the design did not meet all the requirements. Once the magnitudes of the various component values were determined, the closest real component values with SPICE models were used to replace the most significant components of the converter. As a result, the final design differs from the initial estimations in certain areas.

The final simulation schematic of the primary side is shown in Figure 20. The input voltage source V1 is seen on the top left corner and the power transformer TR1 on the top right corner, while the current sense transformer T1 is placed between them. The clamp circuit with the auxiliary switch Q1 and the primary switch Q5 are located underneath them. As seen in the figure, the clamp capacitor U4 value is increased from the calculated value of 3.6 nF to the value of 22 nF. It was found that the efficiency improved by approximately 2 % when the larger clamp capacitance was used. The model of the PWM controller UCC2891, with its 16 pins and their corresponding circuitry, is located on the bottom side of the schematic. The pins of the controller are briefly explained in Table 2. Also, the optocoupler's pull-up resistor R13 was increased from 2.4 kΩ to 4.7 kΩ to improve overshoot and undershoot performance.

Table 2. The pin descriptions of the PWM controller UCC2891

Name	Input/Output	Description
AUX	Output	Gate drive of the auxiliary switch.
CS	Input	Peak current sensing for the current mode control.
FB	Input	Input for the error signal feedback.
GND	-	Analog ground for the PWM control circuitry.
LINEUV	Input	Undervoltage threshold for the input. Enable/disable the outputs based on the input voltage level.
OUT	Output	Gate drive of the main switch.
PGND	-	Serves as the current return for the high-current outputs AUX and OUT.
RSLOPE	Input	Slope compensation for the current mode control circuitry.
RDEL	Input	Used for adjusting the turn-on delay of the gate drives.
RTOFF	Input	Used for adjusting the discharging of the internal timing capacitor.
RTON	Input	Used for adjusting the charging of the internal timing capacitor.
SS_SD	Input	Used for adjusting the soft-start interval of the controller.
SYNC	Input	Used for synchronizing the internal oscillator with an external clock.
VDD	Input	The power supply of the device.
VIN	Input	Used for initiating the operation of the controller.
VREF	Output	5 V reference for the internal logic.

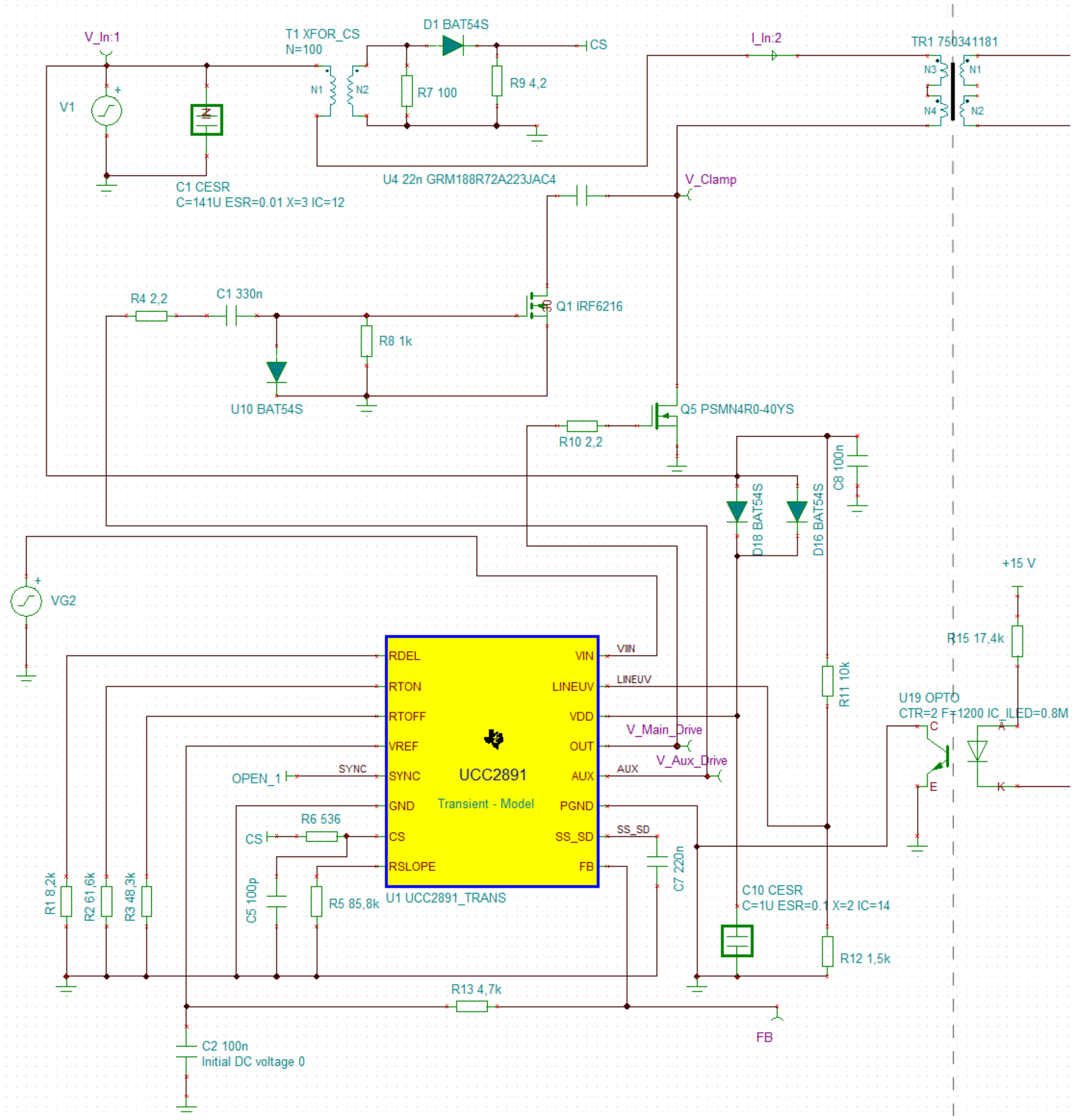


Figure 20. Primary side of the new converter.

The secondary side of the converter is shown in Figure 21. The secondary side consists of rectifier diodes D2, D5, D12, D13, D14, and D15, following the output inductor L1. According to the prior calculations, WE HCF 7443634700 is chosen as the output inductor since it has an inductance of 47 µH and a current rating of 12 A [33]. After the output inductor, there are also the filter capacitors U5 and U2 and resistors R23, R24, R29, and R30 in parallel with the output. The output capacitance was increased from the calculated value because the ripple voltage of the converter was fairly high when a 21 µF capacitor was used. The final value of 846 µF is based on the IPPSuM 1.0, which has one larger 470 µF and eight smaller 47 µF capacitors. Although a smaller increment in the output capacitance might have been enough to smooth out the ripple voltage, the final large output capacitance is justified by the fact that it serves as a buffer for the linear stage, which is ultimately the output that the user sees. The models used in the simulations are Murata's KCM55WR7YA476MH01 ceramic capacitor and Panasonic's EEHZS1E471P hybrid aluminum electrolytic capacitor [34][35].

The feedback circuitry can be seen at the bottom of the schematic, monitoring the output, and bringing the error signal to the primary side through the optocoupler U19. As can be seen, the compensation loop had to be resized since the output inductance and capacitance values were changed. The voltage generator VG_Ref models the reference voltage coming from the MCU and linear side of the IPPSuM. By changing the voltage between 0.6 V and 2.25 V, the converter's output voltage can be adjusted. VS1 serves as a voltage supply for the error amplifier U13 and optocoupler U19. Finally, the voltage pins and current arrows named in purple are measurement points for the simulations.

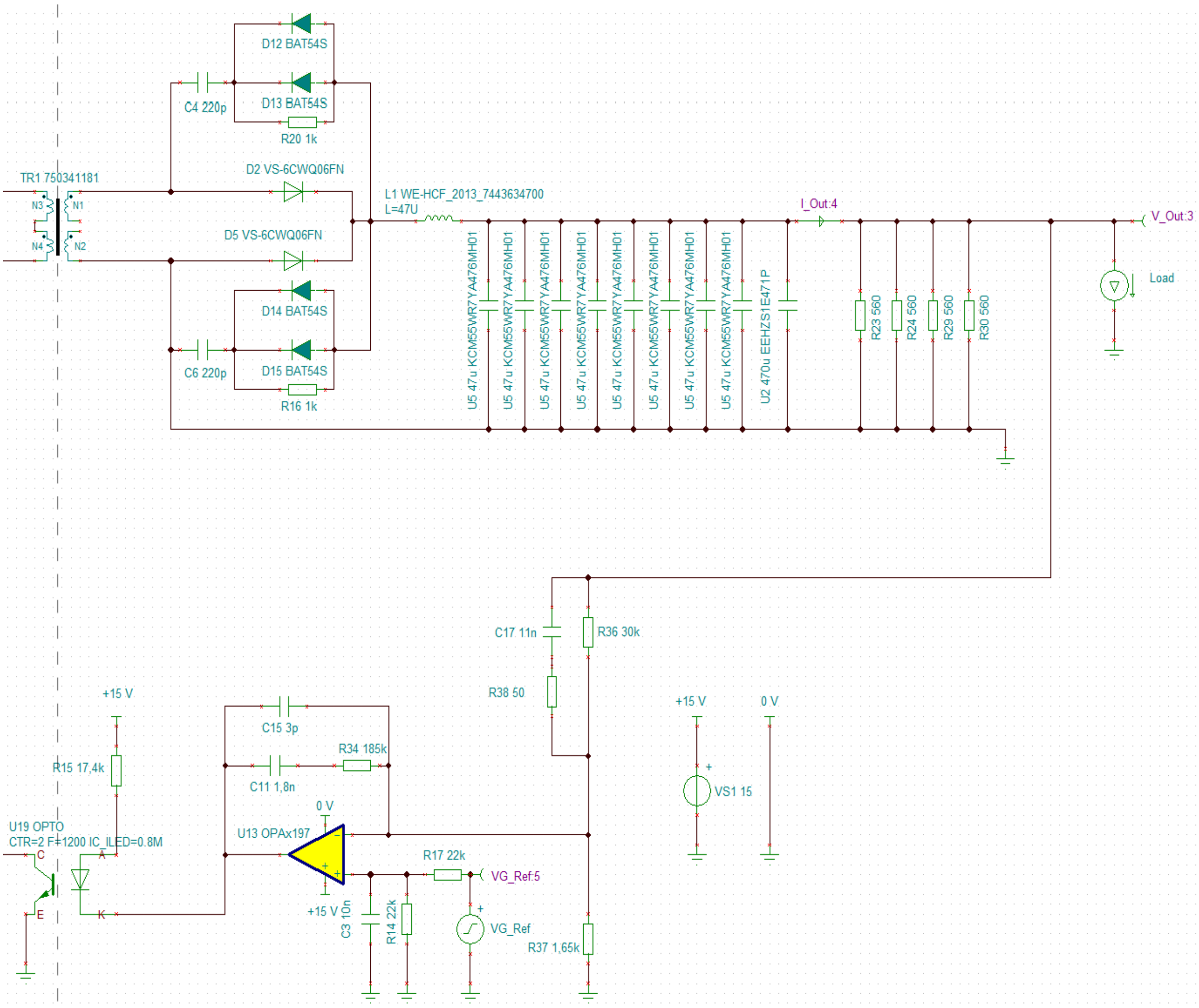


Figure 21. Secondary side of the new converter.

4.1 Output Voltage Range

A couple of details about the Tina-Ti simulation diagram should be noted before looking at the simulation results. Unfortunately, the simulation diagram does not fully display the cursors. In order to maintain clarity, it should be mentioned that the A-cursor is represented by the red line and the B-cursor by the blue line. Another issue is the cursor box, which shows the cursors' delta values. Since it cannot be enlarged, explanatory texts with a larger font size can be seen below the box when it is found to be necessary. Finally, the simulation curves whose labels are circled in red are the focus of interest.

As stated before, the minimum output voltage of the switch-mode stage must be below 6 V, and the maximum output voltage must be close to 21.5 V. In Figure 22, the VG_Ref is swept from 0.6 V to 2.25 V when the maximum load of 5 A is used to illustrate the whole output voltage range of the new converter. As seen in the figure, the output voltage of the converter V_Out is 5.75 V, while the reference voltage V_Ref is 600 mV (red A-cursor). When the VG_Ref rises to the value of 2.25 V, the output voltage V_Out stabilizes to the value of 21.58 V (blue B-cursor). As a conclusion the output voltages are close enough to the target voltages and can be managed by the linear stage.

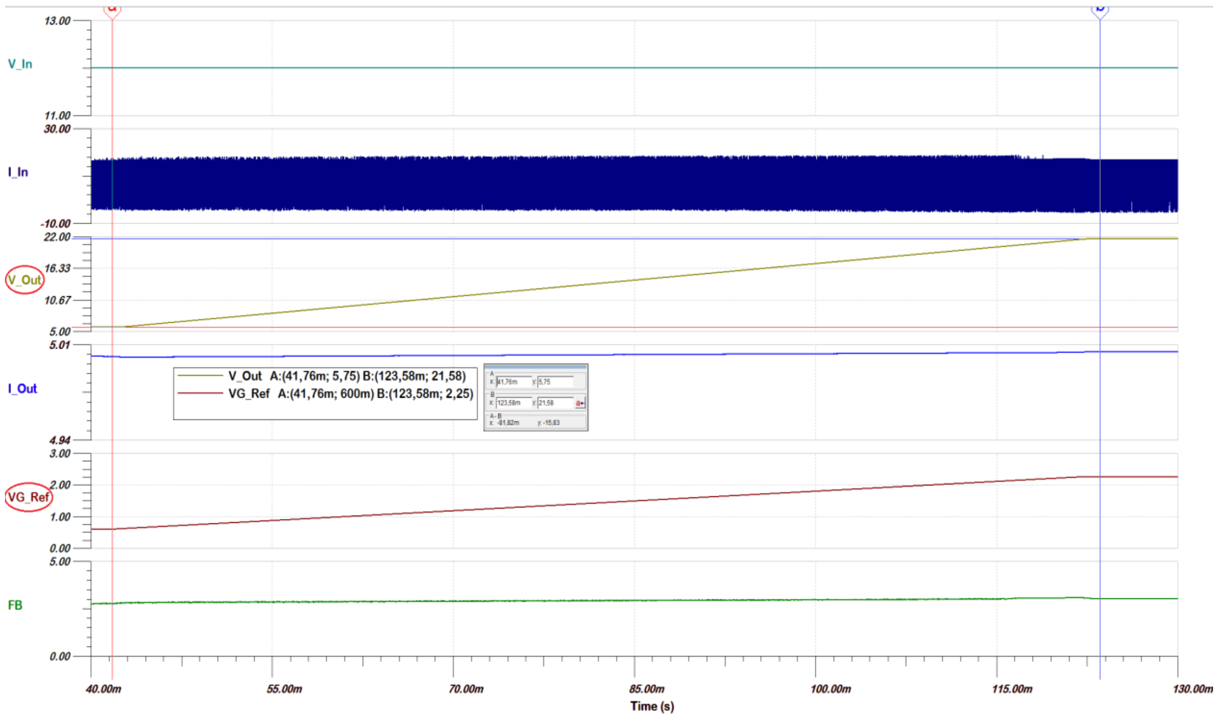


Figure 22. Full output voltage range of the updated converter.

4.2 Output Ripple Voltage

The ripple voltage means the fluctuation of the voltage. As the maximum output ripple voltage of the switch-mode stage is not specified, the total ripple voltage of the IPPSuM 1.0 is still specified to below 30 mV. Therefore, it is necessary that the switch-mode stage of the IPPSuM 2.0 has a ripple voltage well below 30 mV in order to have a good margin for the linear stage. The output ripple is simulated using the maximum output voltage of 21.58 V and the maximum load of 5 A.

As seen in Figure 23, the ripple of the output voltage V_Out is below 1 mV and leaves an unnecessary good margin for the rest of the power supply. The result is mainly achieved using the large output capacitance value, meaning it would be possible to reduce the capacitance and still have good ripple voltage results. However, the value was taken from the old converter. This means that the physical size of the capacitors is not the limiting factor and reducing the output capacitance also affects the feedback loop compensation, which in turn would require more optimization. The large output capacitance serve as a buffer for the linear stage as well. Therefore, the large output capacitance is justified, and this leads to a good ripple voltage result, which should be seen as a positive outcome.

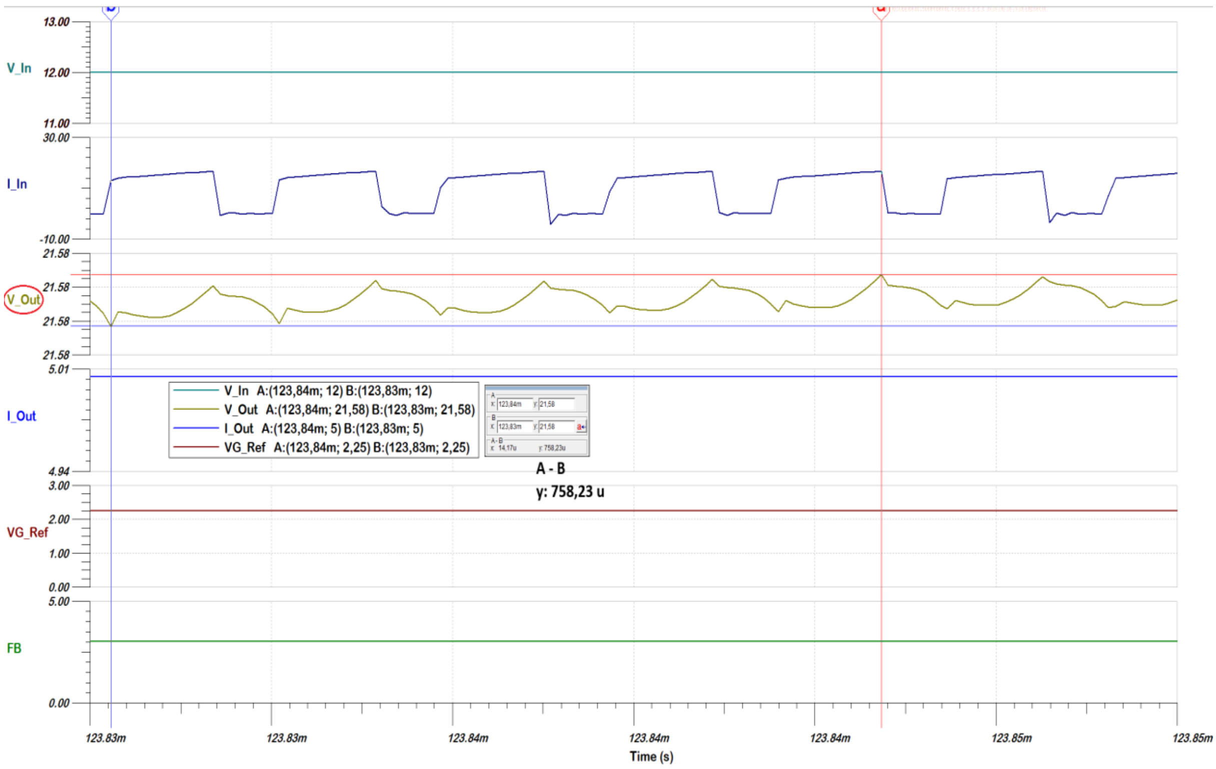


Figure 23. Output voltage ripple at maximum output voltage and the maximum load.

4.3 Load Regulation

Load regulation means the converter's capability to maintain constant output voltage level despite changes in the load. The overshoot and undershoot of the IPPSuM 2.0's converter must be below 100 mV during the load change between 0.2 A and 1.5 A with a slew rate of 0.5 A / µs. In Figure 24, the overshoot is simulated at the maximum output voltage level by reducing the load current at the mentioned slew rate. As seen in the figure, the overshoot of the output voltage V_Out is approximately 55 mV, meaning it is about 0.25 % of the desired output voltage.

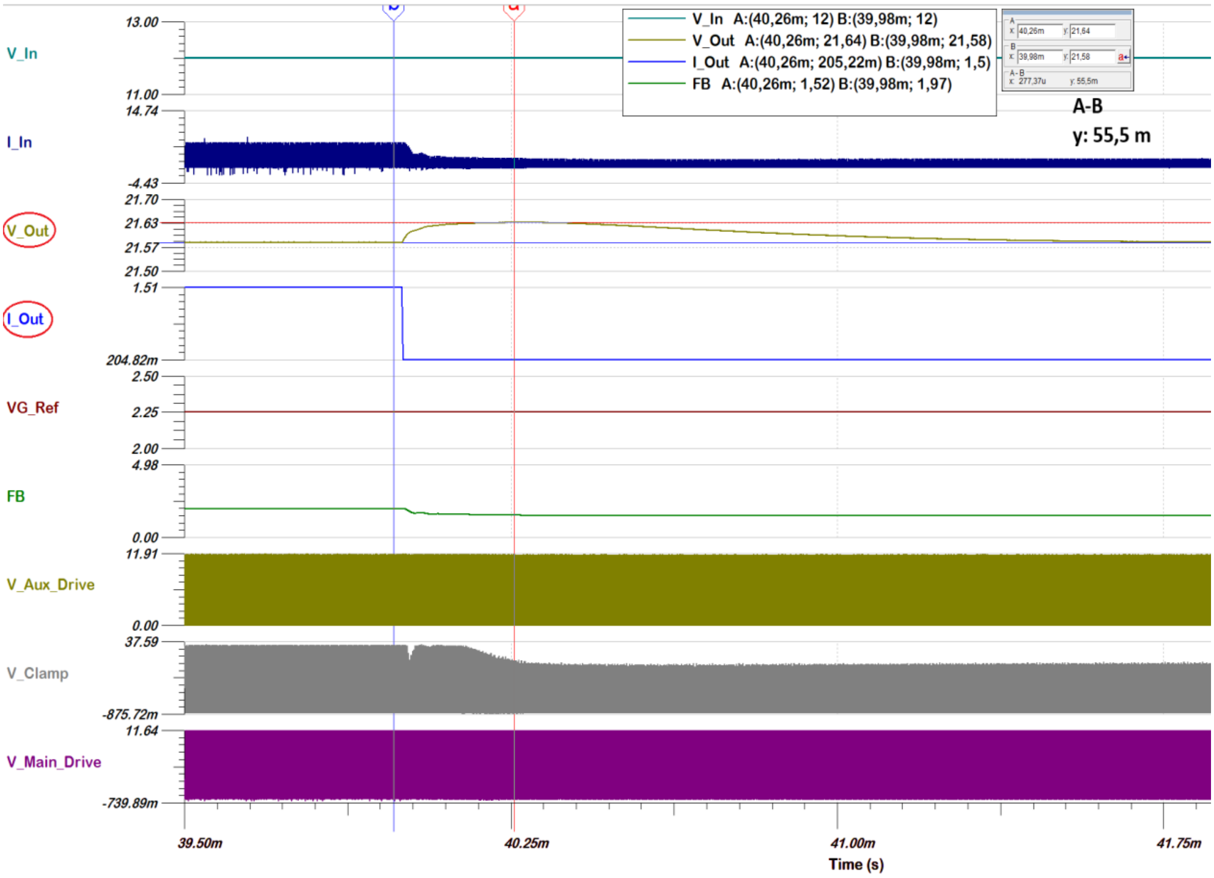


Figure 24. Overshoot during load change from 1.5 A to 0.2 A. Slew rate 0.5 A / µs

In Figure 25, the undershoot is simulated the opposite way compared to the overshoot, meaning the load current is increased at the mentioned slew rate. According to the simulation, the value of the V_Out's undershoot is approximately -59 mV, meaning it is about 0.27 % of the desired output level. In conclusion, both overshoot and undershoot met the requirements greatly.

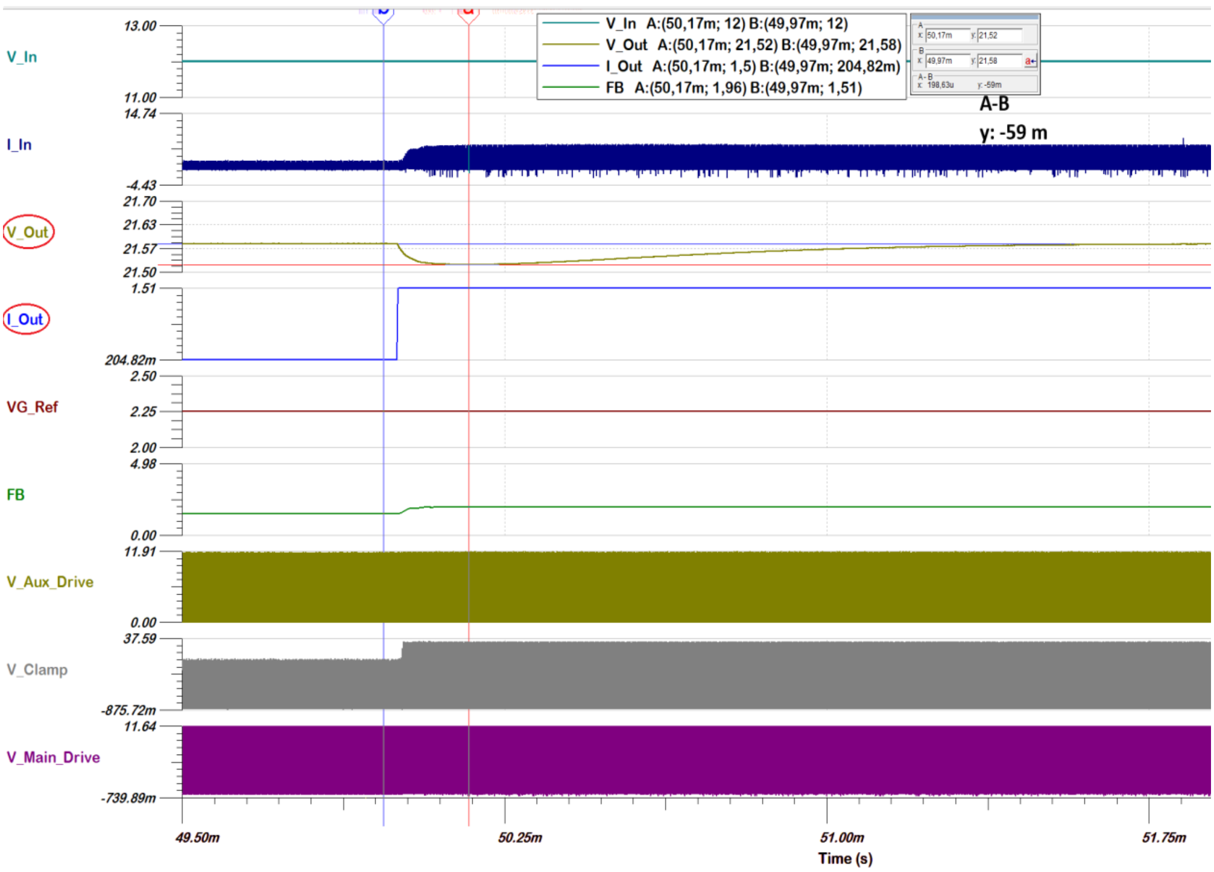


Figure 25. Undershoot during the load change from 0.2 A to 1.5 A. Slew rate 0.5 A / µs.

4.4 Line Regulation

Line regulation means the converter's capability to maintain a steady output voltage level when the input voltage changes. In this case, the output capabilities with the minimum and maximum input voltages of the IPPSuM 2.0's converter are tested. Therefore, the converter should be able to operate properly within the input voltage range of 11.0 V to 13.5 V. The only important values are the final stabilized output voltages because the transition times are not specified. In Figure 26, the line regulation was simulated using the maximum output voltage and the maximum load, as the input voltage level (V_In in Figure 26) was decreased from 13.5 V to 11.0 V. As the figure illustrates, the output voltage (V_Out in Figure 26) requirement of 21–22 V is met by the value of 21.58 V when the input voltage level is 13.5 V. However, when the input voltage level is 11 V, the output voltage falls and eventually stabilizes to the value of 20.7 V, which is slightly under the requirement. As can be seen, in transition, the output voltage falls even lower, but as the recovery time is not absolutely poor, the main focus is the stabilized final value, as stated before.

The output voltage level with the minimum input voltage could be increased by using a higher duty cycle or by increasing the turns ratio of the power transformer. Since the UCC2891 controller is already using its maximum duty cycle, the only viable option is to utilize a different power transformer.

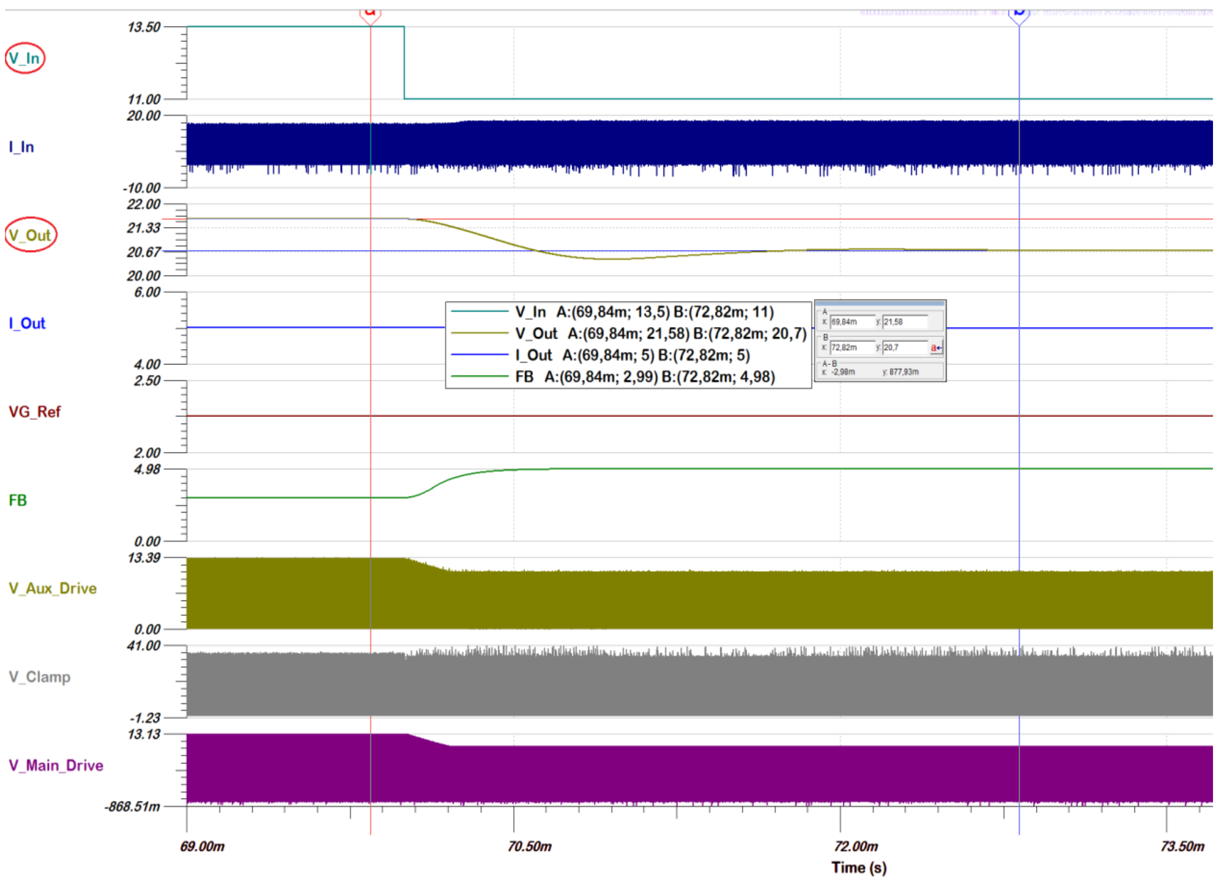


Figure 26. Line regulation of the converter.

4.5 Efficiency

The efficiency requirement of the converter is not specified. However, because of the power supply's small physical size and lack of a large heat dissipation area, it should have a relatively high efficiency. The design was simulated using the maximum, midway and minimum output voltages. All scenarios were also simulated using five different load settings: 100 %, 75 %, 50 %, 25 %, and 10 % of the maximum load. The values of the input and output voltages and currents can be calculated to the respective powers with equation (55):

$$P = U * I, \tag{55}$$

where

P is the power,
U is the voltage and
I is the current.

As the input and output powers are known, the efficiency can be calculated with equation (56):

$$\eta = \frac{P_{OUT}}{P_{IN}} * 100\ \%, \tag{56}$$

where

η is the efficiency,
P_{OUT} is the output power and
P_{IN} is the input power.

In Figure 27, the design is simulated with the maximum output voltage and the maximum load. The boxes in the figure display the values of the voltages (V_In and V_Out in Figure 27) and currents (I_In and I_Out in Figure 27) required to calculate efficiency. The average value is used for the input current, I_In. Using equations (55) and (56), the efficiency of the converter in this scenario was calculated to be 91.8 %.

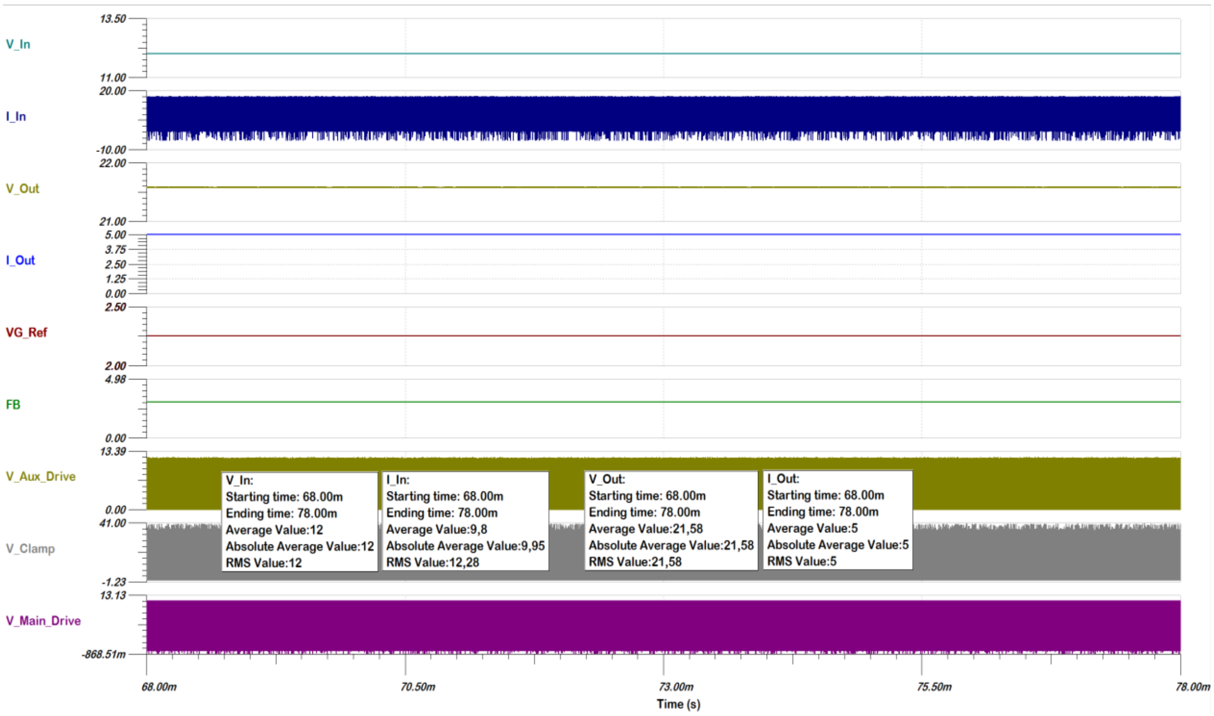


Figure 27. Input and output voltages and currents at the maximum output voltage and the maximum load.

The simulations of other scenarios can be seen in appendices. The simulation results and calculations are compiled into three separate tables, each containing the results for the five load current levels (10 %, 25 %, 50 %, 75 %, and 100 %). The output voltage differs between the tables; Table 3 shows the results at the maximum output voltage (21.58 V), Table 4 shows the results at the midway output voltage (13.67 V), and Table 5 shows the results at the minimum output voltage (5.75 V). Input and output powers P_In and P_Out are calculated with equation (55), and efficiency η with equation (56).

As can be seen in Table 3, the efficiency is between 87.3 and 92.5 % when the output voltage is at its maximum. When the output voltage is between the minimum and the maximum, the efficiency is slightly better, from 90.7 to 93.7 %, as can be seen in Table 4. Table 5 shows the worst results at the minimum output voltage, with the efficiency being between 83.2 and 86.8 %. To conclude, the converter is the most efficient when the output voltage is at the midway value of 13.67 V and the load current is 25 % of the maximum load. On the other hand, the worst efficiency of 83.2 % is calculated at the minimum output voltage with the maximum load.

In conclusion, the efficiency results appear to be comparable to typical switch-mode converter efficiency. However, at low voltage levels, the efficiency is the lowest. It is mainly a result of a voltage drop caused by the output rectifier diodes. As the voltage drop may be considered constant, it affects the output voltage more at low voltage levels compared to higher voltage levels. On the other hand, the efficiency is highest at midway voltages closer to the input voltage. If the output voltage has to be stepped up from the input voltage, more power is required, which in turn increases the conduction losses of the circuit.

Table 3. Efficiency simulation results and calculations at the maximum output voltage

Load [%]	V_In [V]	I_In [A_{AVG}]	V_Out [V]	I_Out [A]	P_In [W]	P_Out [W]	η [%]
10	12.00	1.05	21.58	0.51	12.60	11.01	87.3
25	12.00	2.46	21.58	1.25	29.52	27.00	91.4
50	12.00	4.86	21.58	2.50	58.32	53.95	92.5
75	12.00	7.31	21.58	3.75	87.72	80.93	92.3
100	12.00	9.80	21.58	5.00	117.70	107.90	91.8

Table 4. Efficiency simulation results and calculations at the midway output voltage

Load [%]	V_In [V]	I_In [A_{AVG}]	V_Out [V]	I_Out [A]	P_In [W]	P_Out [W]	η [%]
10	12.00	0.62	13.67	0.50	7.44	6.84	91.9
25	12.00	1.52	13.67	1.25	18.24	17.09	93.7
50	12.00	3.05	13.67	2.50	36.60	34.18	93.4
75	12.00	4.58	13.67	3.75	54.96	51.26	93.3
100	12.00	6.28	13.67	5.00	75.36	68.35	90.7

Table 5. Efficiency simulation results and calculations at the minimum output voltage

Load [%]	V_In [V]	I_In [A_{AVG}]	V_Out [V]	I_Out [A]	P_In [W]	P_Out [W]	η [%]
10	12.00	0.29	5.75	0.50	3.44	2.88	83.5
25	12.00	0.69	5.75	1.25	8.30	7.19	86.6
50	12.00	1.38	5.75	2.50	16.56	14.38	86.8
75	12.00	2.09	5.75	3.75	25.08	21.56	86.0
100	12.00	2.88	5.75	5.00	34.56	28.75	83.2

5 DISCUSSION

This thesis covered the design and simulation of the DC-DC active clamp forward converter, with the purpose of upgrading the existing converter to have wider output voltage and power ranges. The output voltage of the converter was increased from the 5.18 – 11.45 V range to the 5.75 – 21.58 V range. It was relatively successful, given that the initial target range was 5.5 – 21.5 V. The converter was also capable of handling the maximum load current of 5 A with the full output voltage range and therefore satisfied the maximum power requirement.

The design also met all other requirements well, except the line regulation with the minimum input voltage of 11 V. It was found that the output voltage failed to reach the required level when the minimum input voltage was used. The specifications could be reviewed again to determine whether the minimum voltage is necessary to be 11 V. The actual minimum input voltage required by the converter to regulate a higher than 21 V output voltage was not simulated, but it is possible that the converter does not require a significant voltage increase to do this. If it was found that 11 V is critical, increasing the turns ratio of the power transformer is the solution since the duty cycle cannot be increased any further. Also, the line regulation during the transition times got little attention since there was no specification for it and the main focus was on the final stabilized output voltage values. Therefore, it should be verified more thoroughly after the final desired values are figured out and the design meets these specifications.

The maximum and minimum output voltages are close enough to the target values and should be managed by the linear stage. However, if tuning is required, the final optimization could be done to the reference voltage provided by the MCU and linear stage.

When it comes to the output ripple voltage, the results were on a great level and therefore leave almost the whole 30 mV requirement as a margin for the rest of the power supply. On the other hand, there is room for decreasing the output capacitance, which has the most impact on the output voltage ripple. This would allow the converter to respond faster to load transients as the settling time is quicker. It would also allow the design of a smaller power supply in terms of physical size. However, the total ESR value of the output capacitance must be considered. For example, too low ESR might cause the converter to oscillate and could make the compensation of the feedback loop difficult. After all, the chosen output capacitance value was copied from the old IPPSuM 1.0, meaning the physical size should not be an issue. In addition, the large output capacitance helps to maintain the output voltage steady due to sudden changes in load current.

Also, the load regulation results met the specifications well. Nevertheless, in order to obtain comparable results, the load regulation simulations were only done with fairly low load currents according to the specifications of the old IPPSuM 1.0. In order to get a better picture of the performance of the converter, it could be beneficial to do the simulations with higher load currents and a wider load current range as well.

Efficiency was not a key priority in the design. Yet, the efficiency results were good and in line with the typical switch-mode converter's level. However, efficiency was clearly at its worst when the lowest output voltage was used. It could be improved by using synchronous rectification on the secondary side. By replacing the output rectifier diodes with MOSFETs, the voltage drop caused by diodes can be avoided. This affects efficiency, especially at low voltage levels where the voltage drop across a diode is significant in relation to the output voltage level.

The designed converter was based on the old converter and strictly followed its specifications. However, there are also a couple of other methods for improving efficiency if the power supply were to be totally redesigned, with efficiency being the main priority. The first thing is to consider utilizing a higher input voltage for the power supply. As the input power remains constant, the input current drawn is lower, which should directly reduce the conduction losses. Lowering the switching frequency might also improve efficiency (and reduce EMI), resulting in lower switching losses, but at the cost of larger components and slower transient response.

Finally, the feedback circuit was greatly simplified from its original design. As a result, it requires fewer components, which saves space on the circuit board and also lowers component costs. However, little time was spent optimizing the feedback circuit, so it could potentially be improved further.

The new DC-DC converter allows the entire battery emulator power supply to be upgraded, increasing the number of possible devices that can be tested with it in the future. These devices that could benefit from the increased voltage range are, for example, certain battery-powered cordless tools and Bluetooth speakers operating at 18 V. Furthermore, the upgraded power supply would challenge the few competing products on the market. The increased voltage range would be comparable, if not identical, to others [36][37]. Also, the current sink capability appears to be in line with others. While it would lose little on the power sourcing capability, the IPPSuM's compact size will certainly be an advantage, as it is almost half the size compared to others. That being said, the final comparison cannot be performed until the entire power supply has been specified and designed.

6 SUMMARY

IPPSuM, a battery emulator power supply, has been designed for testing battery-powered devices. The power transfer of the compact power supply is achieved with a DC-DC switch-mode converter, followed by a linear regulator. However, the problem with the power supply has been the relatively low output voltage and power, which limit the number of devices to be tested. In order to upgrade the power supply to higher voltage levels, the necessary step was to upgrade the DC-DC converter. Hence, the purpose of this work was to design and upgrade the power supply's DC-DC switch-mode converter for wider voltage and power ranges.

Thesis started with basic theory about DC-DC switch-mode converters in order to better understand their principles. The basic theory was followed by the introduction of different switch-mode converter topologies. The topologies that were considered suitable for the small form factor, low power level power supply, were discussed more in detail.

After the theory section, the IPPSuM and the new converter requirements were presented. The output voltage had to be increased from the 5.18 V – 11.45 V range to the target range of 5.5 V – 21.5 V. Based on the old IPPSuM 1.0 and the new requirements, the active clamp forward topology was found to be the ideal approach for the new converter. The topology was found to be capable of operating at a high duty cycle, resulting in higher output voltages while reducing voltage stress on the switches. Furthermore, it was used in the old IPPSuM 1.0, meaning it appeared to be the most straightforward way to upgrade the old converter to a higher voltage level.

Topology selection was followed by the design section, which presented the component sizing calculations. The calculations were done according to the datasheet of the chosen PWM switch-mode controller and other examples from Texas Instruments. Finally, after the component sizing, the performance of the converter was simulated using Tina-Ti software. The design specifications and requirements were based on the old IPPSuM 1.0. The simulation results showed that the voltage range was increased relatively successfully to the 5.75 V – 21.58 V range. The converter was also capable of handling the maximum load current of 5 A with the full output voltage range and therefore satisfied the maximum power requirement. Also, the design met all other specifications well, except the line regulation requirement with the minimum input voltage of 11 V. It was found that the power transformer turns ratio has to be increased in order to solve this problem. In the end, it was discussed that it would be beneficial to do more simulations to ensure the best performance of the converter, as this thesis could not cover everything.

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[36] Datasheet Keithley Series 2281S Battery Simulator and Precision DC Power Supply.

[37] Datasheet Keysight E36731A Battery Emulator and Profiler.

8 APPENDICES

Appendix 1 Input and output voltages and currents at the maximum output voltage and 75 % of the maximum load

Appendix 2 Input and output voltages and currents at the maximum output voltage and 50 % of the maximum load

Appendix 3 Input and output voltages and currents at the maximum output voltage and 25 % of the maximum load

Appendix 4 Input and output voltages and currents at the maximum output voltage and 10 % of the maximum load

Appendix 5 Input and output voltages and currents at the midway output voltage and 100 % of the maximum load

Appendix 6 Input and output voltages and currents at the midway output voltage and 75 % of the maximum load

Appendix 7 Input and output voltages and currents at the midway output voltage and 50 % of the maximum load

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Appendix 10 Input and output voltages and currents at the minimum output voltage and 100 % of the maximum load

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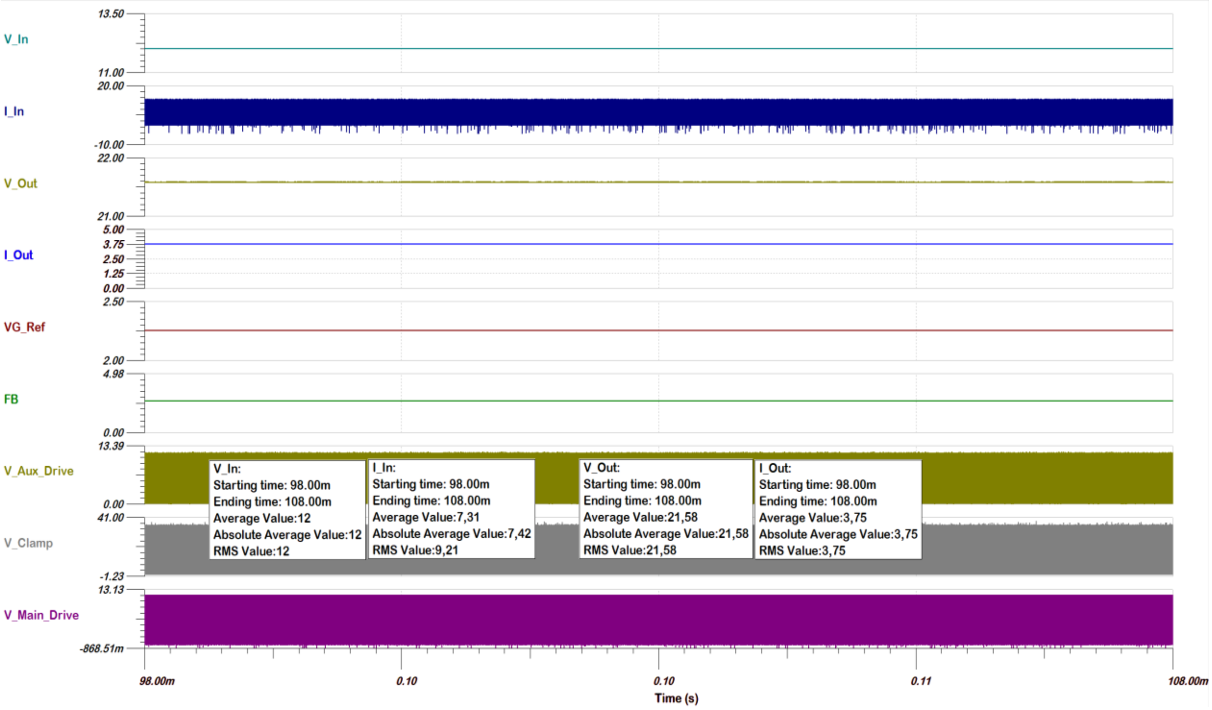
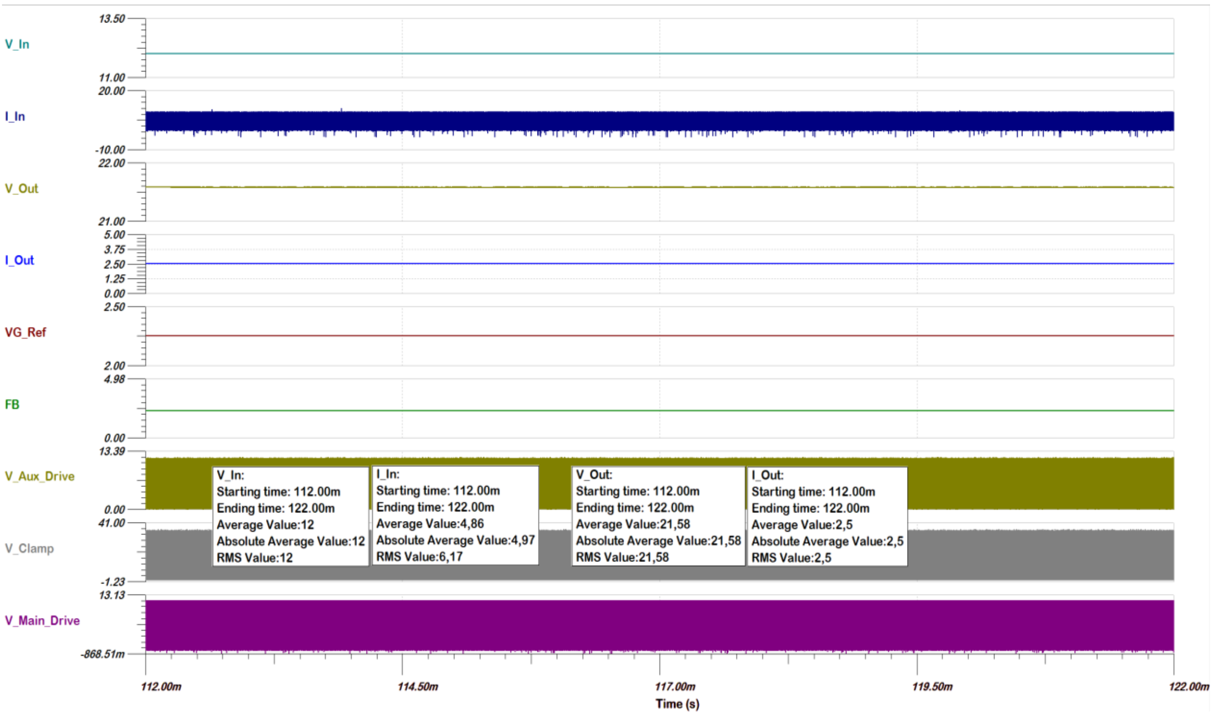
Appendix 13 Input and output voltages and currents at the minimum output voltage and 25 % of the maximum load

Appendix 14 Input and output voltages and currents at the minimum output voltage and 10 % of the maximum load

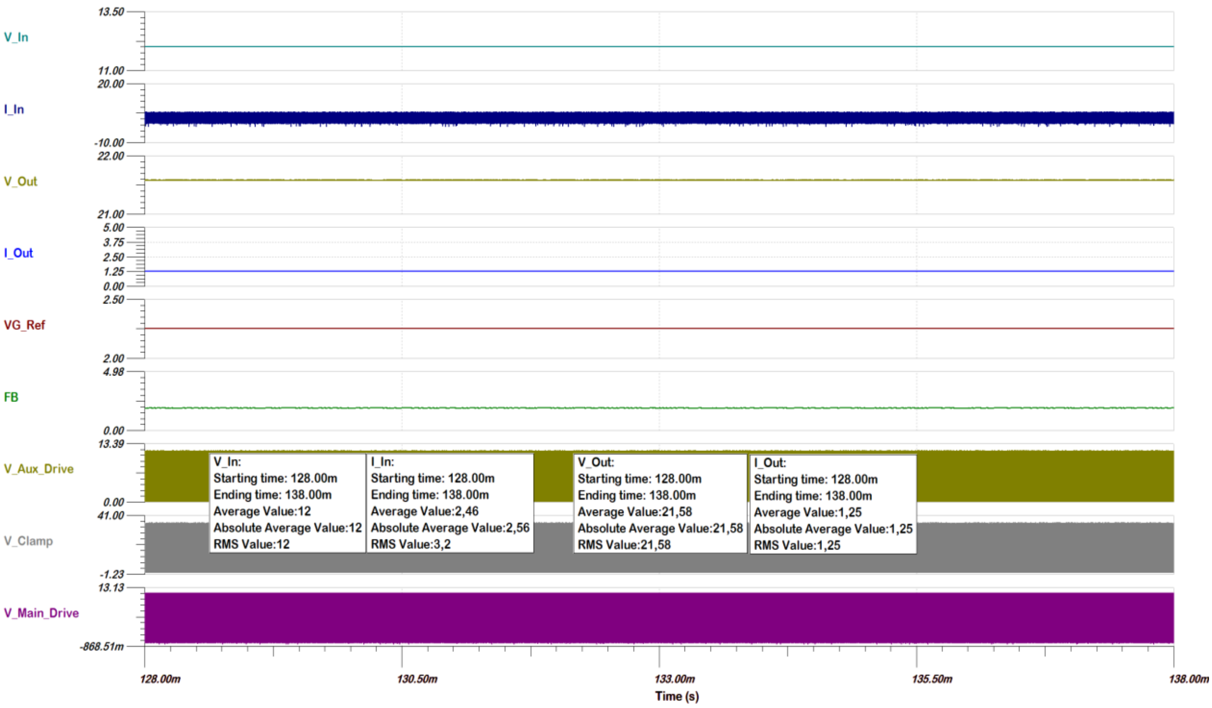
Appendix 1 Input and output voltages and currents at the maximum output voltage and 75 % of the maximum load



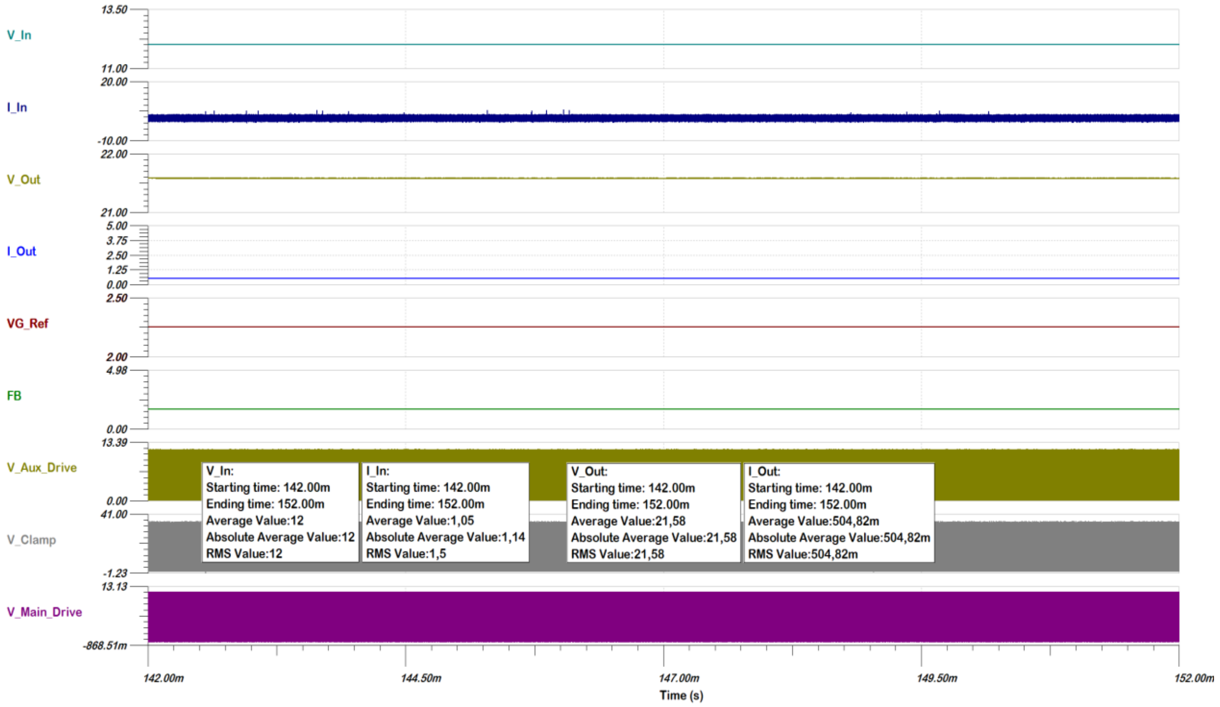
Appendix 2 Input and output voltages and currents at the maximum output voltage and 50 % of the maximum load



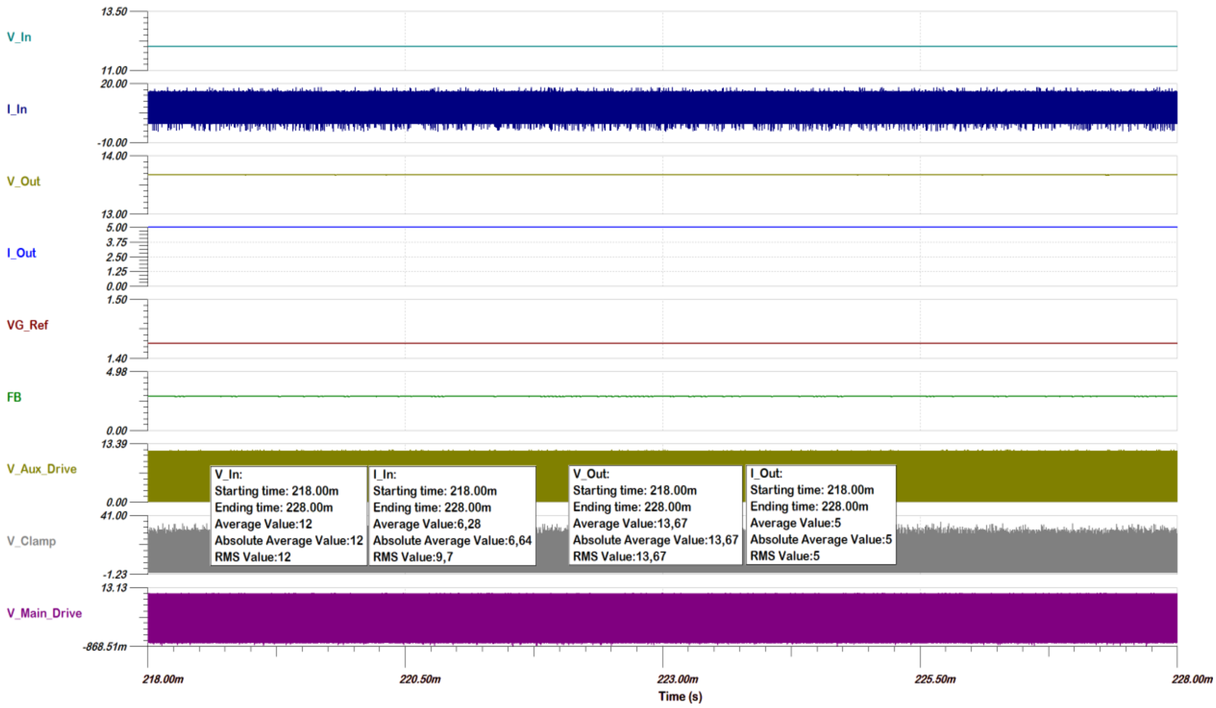
Appendix 3 Input and output voltages and currents at the maximum output voltage and 25 % of the maximum load



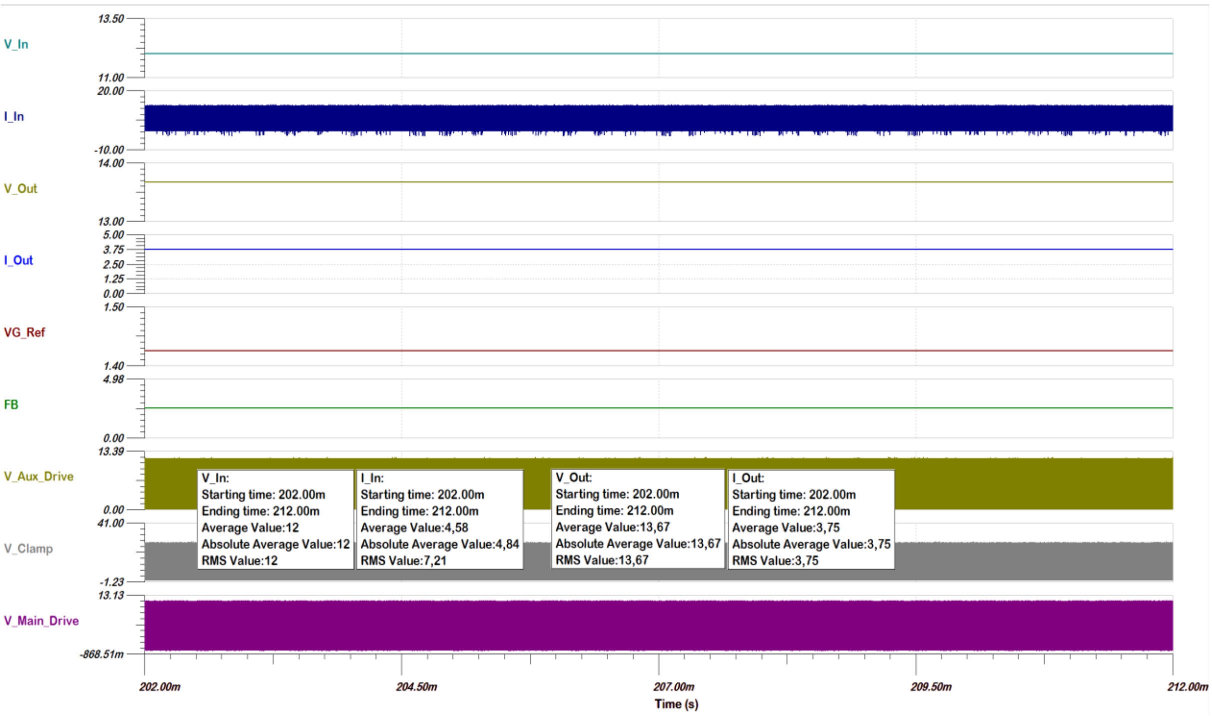
Appendix 4 Input and output voltages and currents at the maximum output voltage and 10 % of the maximum load



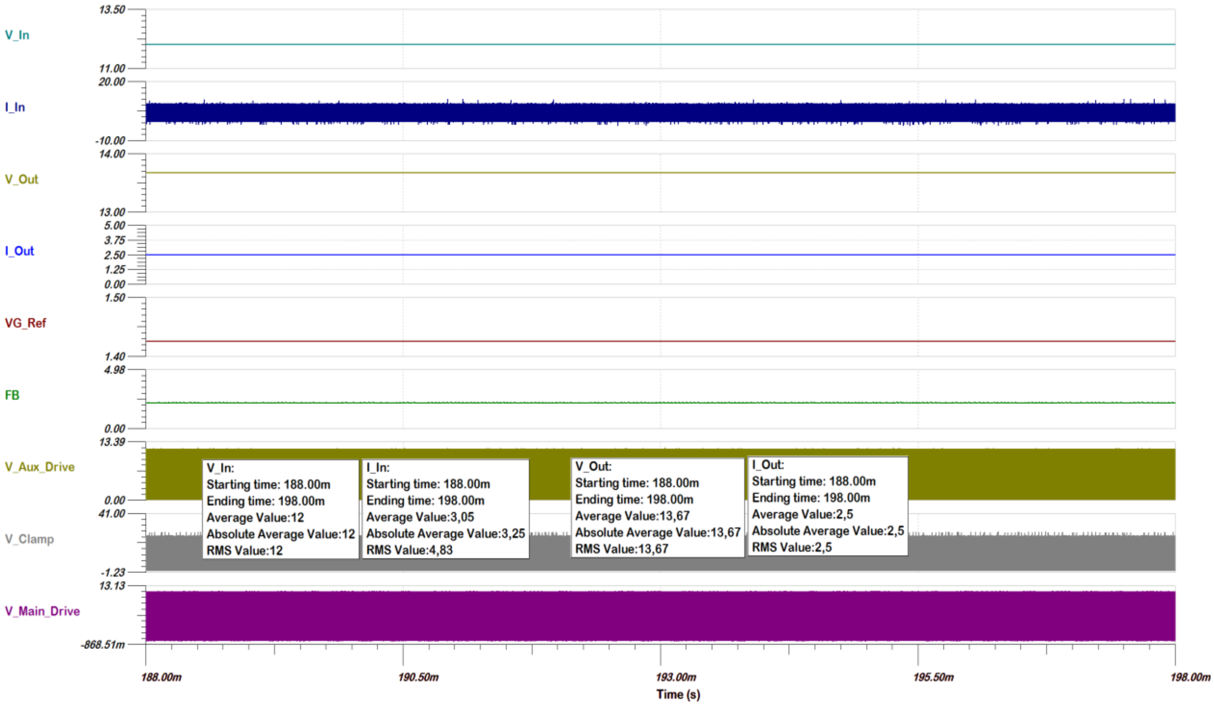
Appendix 5 Input and output voltages and currents at the midway output voltage and 100 % of the maximum load



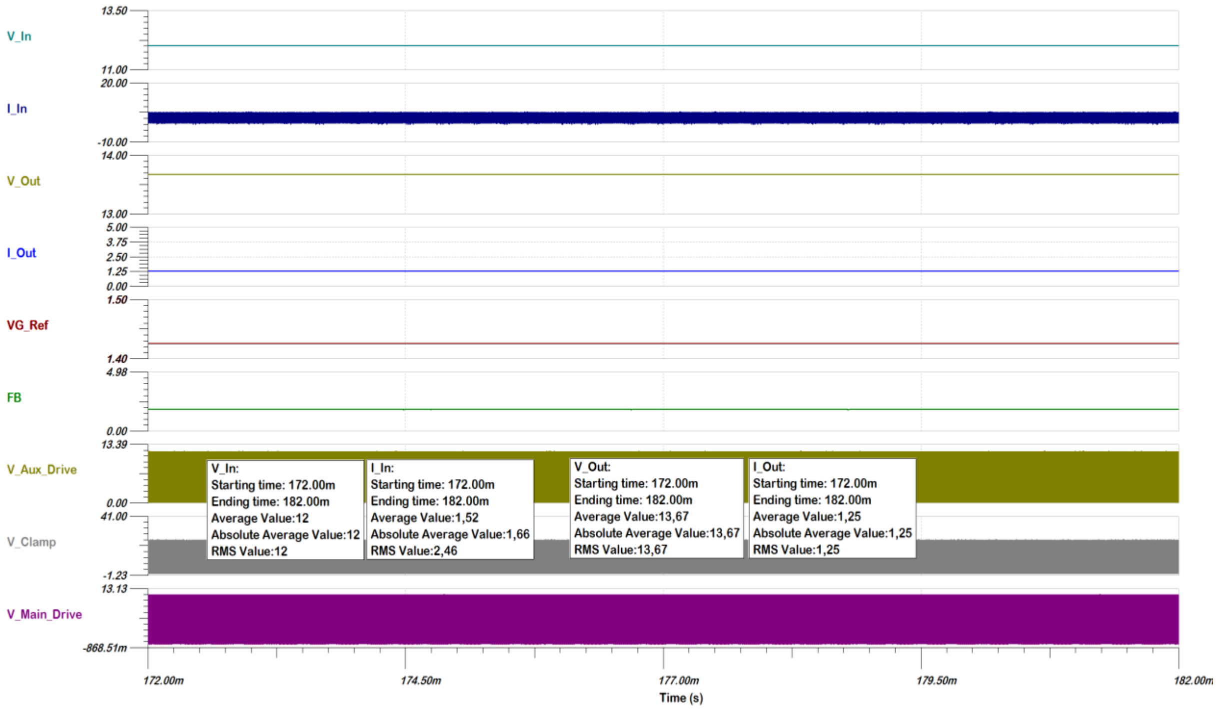
Appendix 6 Input and output voltages and currents at the midway output voltage and 75 % of the maximum load



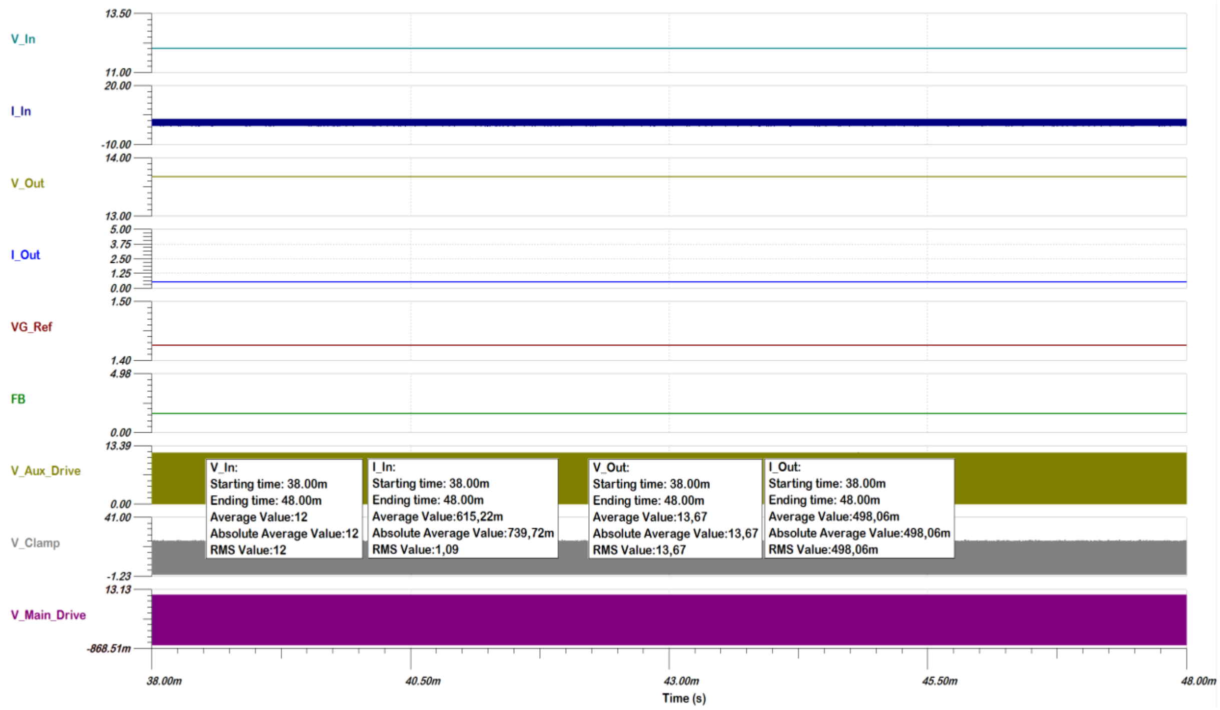
Appendix 7 Input and output voltages and currents at the midway output voltage and 50 % of the maximum load



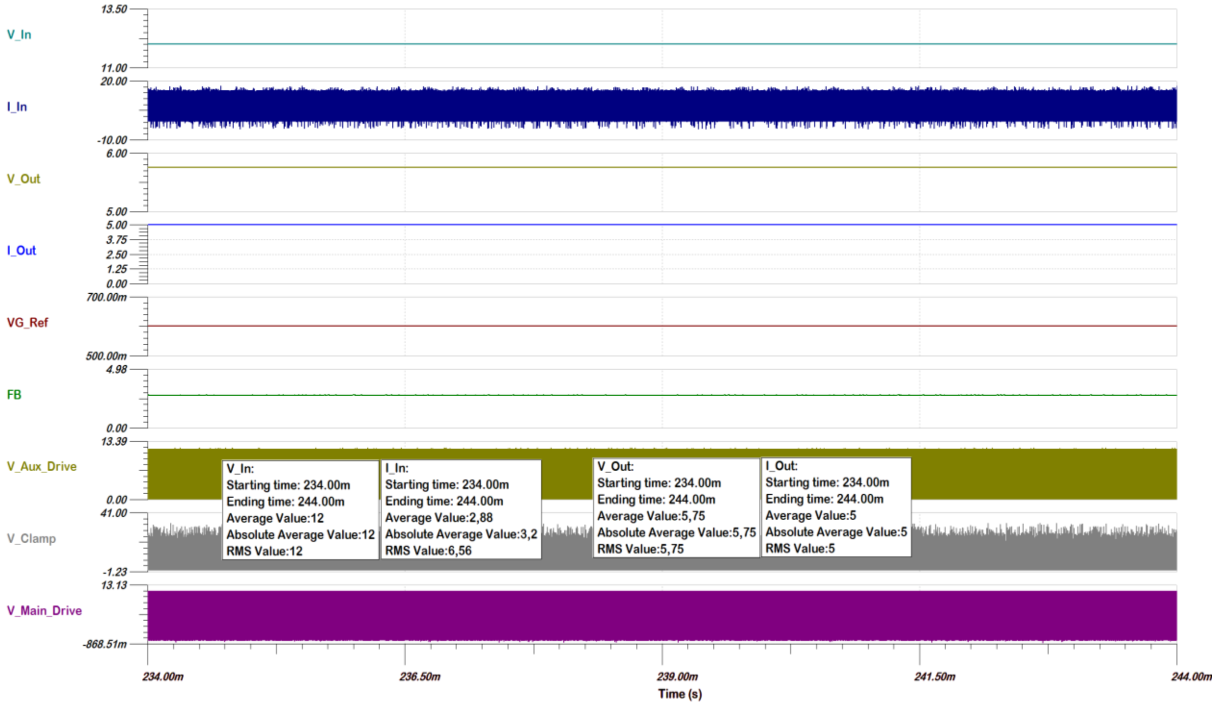
Appendix 8 Input and output voltages and currents at the midway output voltage and 25 % of the maximum load



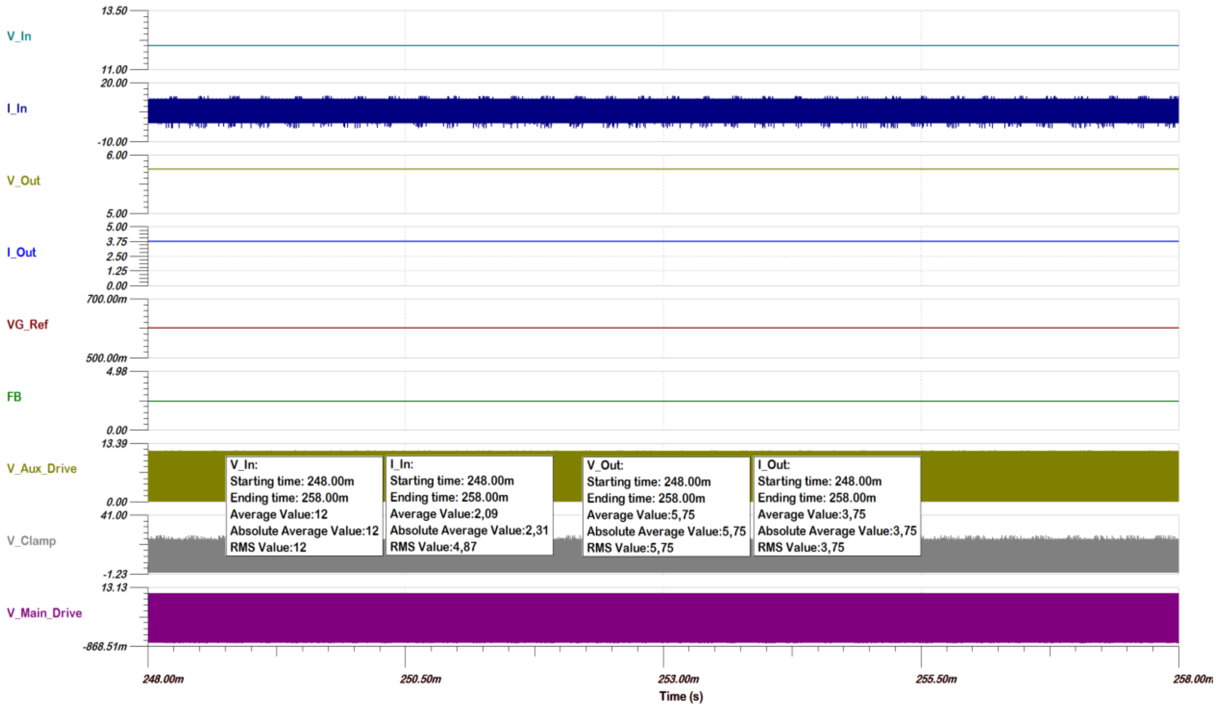
Appendix 9 Input and output voltages and currents at the midway output voltage and 10 % of the maximum load



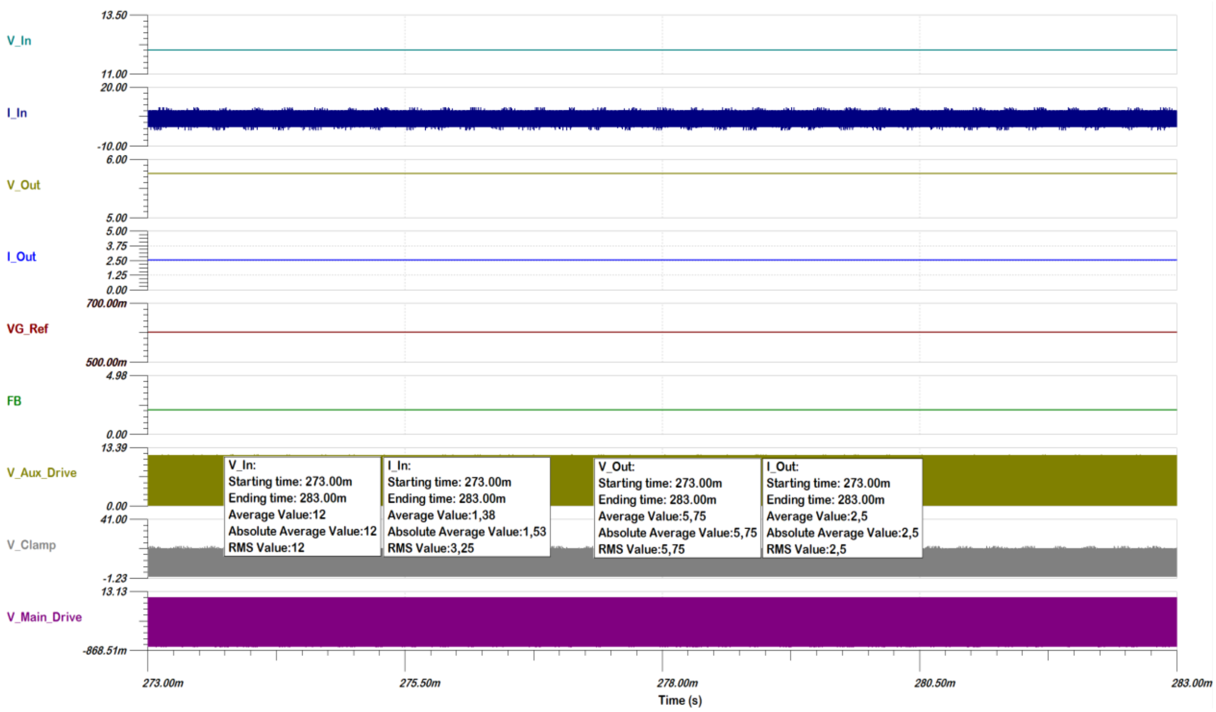
Appendix 10 Input and output voltages and currents at the minimum output voltage and 100 % of the maximum load



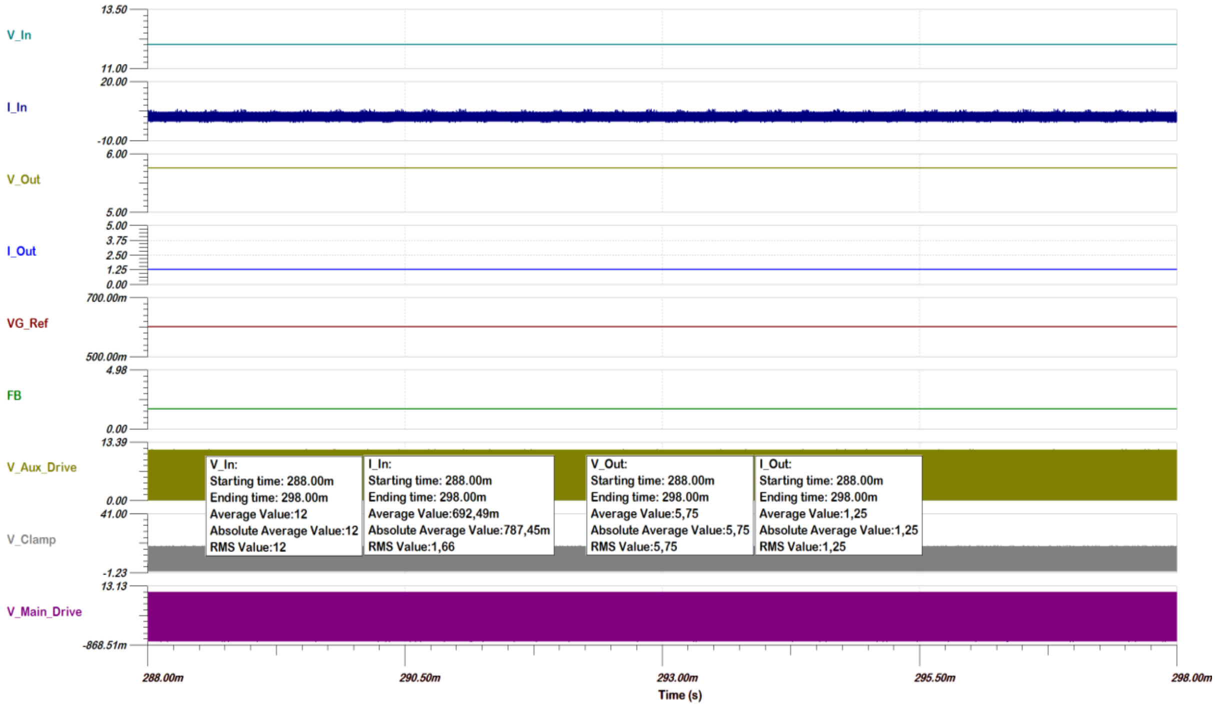
Appendix 11 Input and output voltages and currents at the minimum output voltage and 75 % of the maximum load



Appendix 12 Input and output voltages and currents at the minimum output voltage and 50 % of the maximum load



Appendix 13 Input and output voltages and currents at the minimum output voltage and 25 % of the maximum load



Appendix 14 Input and output voltages and currents at the minimum output voltage and 10 % of the maximum load